

Stablecoins as Digital Havens: Cryptocurrencies in Conflict and Sanctioned Economies

Abstract

This paper examines the increased attention and resulting engagement with digital assets among Russian users following the 2022 invasion of Ukraine and the subsequent imposition of economic and financial sanctions. We document a surge in interest from Russian retail investors in cryptocurrencies, particularly in USD-backed stablecoin Tether. This higher interest corresponds to observable shifts in blockchain activity, most notably an increase in stablecoin usage among smaller retail users. Therefore, our findings suggest that stablecoins can act as digital havens for retail users in conflict and sanctioned economies, enabling users to circumvent financial sanctions and protect the value of their assets.

Keywords: Stablecoins; Sanctions; Blockchain; Russia-Ukraine War; Safe Haven Assets.

1. Introduction

The invasion of Ukraine on 24 February 2022 led to a series of international sanctions aimed at isolating Russia economically and financially. The 27 member states of the EU, the UK and the US, as well as dozens of other countries, imposed a combination of financial, energy sector related, and individual-based sanctions, along with trade and transportation restrictions, and other measures such as media and technological limitations (Girardone, 2022). Approximately \$350 billion or 70% of Russia’s central bank assets¹ have been frozen, with several commercial banks excluded from the SWIFT international payment system. These sanctions initially disrupted the country’s financial system, driving the Russian economy to four consecutive quarters of negative economic growth. However, later on they have proved less effective than anticipated, with Russia shifting to a war economy and returning to pre-sanction growth levels only a year after the invasion.²

In this paper, we examine whether Russian users’ interest in digital assets has increased as a result of these post-invasion imposed international sanctions and whether the heightened interest has focussed on cryptocurrencies in general or stablecoins more specifically. Digital assets rely on settlement systems that cannot (and have not been) affected by sanctions given the pseudonymous privacy they offer. Hence, they can act as a digital haven to circumvent financial sanctions. To explore the shift in Russian retail investors’ interest in digital assets triggered by these events, we use Google Trends data for keywords related to cryptocurrencies and stablecoins and apply a Differences-in-Differences (DiD) analysis, complemented by a suite of robustness tests. In addition, we investigate the impact of increased public attention on the adoption and usage levels of digital assets using blockchain on-chain metrics.

Our findings suggest that Russian retail investors’ interest in cryptocurrencies and stablecoins in particular has increased during the conflict with Ukraine. Stablecoins (and all digital assets) are unaffected by imposed financial sanctions and can be used as a digital haven for hedging financial and geopolitical risks, enabling users to preserve their wealth. This heightened interest results in a significant and enduring divergence in retail investors’ attention to stablecoins between

¹Source: <https://tradingeconomics.com/russia/foreign-exchange-reserves>

²<https://www.bbc.co.uk/news/world-europe-60125659>

Russia and several countries used as control groups. Additional analysis indicates that this surge in interest aligns with substantial shifts in key on-chain metrics, including increased activity levels and adoption rates in US dollar-backed stablecoin Tether. In particular, we observe an increase in usage among smaller retail users on Tron blockchain. This suggests, in addition to our previous analysis, that not only there is heightened interest in stablecoins, but that this also translates into a higher number of addresses with smaller holdings of stablecoins since the war in Ukraine began.

Our study relates and contributes to several areas of the literature. First, it relates to past research that identifies that negative news about unexpected and unanticipated events can lead to an abnormal increase in the demand for safe haven assets, as investors seek to mitigate exposure to high volatility and negative returns (Barber and Odean, 2008, Cai and Zhao, 2024, Jagolinzer, 2024, Merkley et al., 2024). The increased demand for safe haven assets is often driven by the behavior of retail investors (Barber et al., 2009, 2022). Commodities such as gold,³ government bonds (e.g., T-bills), digital assets (e.g., cryptocurrencies), and, most recently, stablecoins have been examined as potential safe havens during periods of financial turmoil and market turbulence. Past literature suggests that investors turn to safer assets, such as T-Bills, during these periods (see, for instance, Nagel (2016), Adrian et al. (2019) and He et al. (2019), Cieslak and Vissing-Jorgensen (2021)) and that cryptocurrency investors are not very different from traditional sophisticated investors (Auer and Tercero-Lucas, 2022).⁴ Baur and McDermott (2010) find that gold has served as a safe haven during periods of crises and stock market downturns. Existing research provides mixed findings on the use of cryptocurrencies as safe havens (Yae and Tian, 2024, Zhang et al., 2021, Urquhart and Zhang, 2019, Baur et al., 2018). Corbet et al. (2020a), Conlon et al. (2020), Wen et al. (2022) and Ji et al. (2020) find that Bitcoin did not serve as a safe haven during the COVID-19 pandemic. Goodell and Goutte (2021) and Conlon et al. (2020) highlight the role of stablecoins as diversifiers. Corbet et al. (2020b) examine a broad set of responses of digital assets to changes in the interest rate of the US Federal Fund. Cong et al. (2024) and Liu and Valcarcel (2024) show that

³See for instance Reboredo (2013), Baur and McDermott (2010), and Baur and Smales (2020).

⁴However, recent evidence suggests that the role of T-Bills as a safe haven asset has changed, particularly during the COVID-19 pandemic. Cashin et al. (2023), Fleckenstein and Longstaff (2024), He et al. (2022), and Haddad et al. (2021) highlight that during the pandemic, the Treasury market experienced stress and illiquidity, raising concerns that its safe haven status may be eroded.

cryptocurrencies might be perceived as an inflation hedge. Although cryptocurrencies, particularly Bitcoin, were initially designed to function as peer-to-peer cash, economic and technological factors have hindered their widespread adoption as mediums of exchange.⁵ On the contrary, stablecoins link their value to and are redeemable for the major legal tender, predominantly the USD, and, despite challenges, are probably better suited as mediums of exchange (Perez Riaza and Gnabo, 2025, Dionysopoulos et al., 2024, Lyons and Viswanath-Natraj, 2023).

Some previous literature has also investigated the impact of broader geopolitical risk, on both traditional assets and cryptocurrencies. For example, Liu and Zhang (2024) find that geopolitical risk has a strong predictive power in traditional currency markets. Wang et al. (2020), Umar et al. (2021), Mokni (2021), and Lucey et al. (2022) find that geopolitical uncertainty increases Bitcoin returns, volatility, and trading volume. Gill-de Albornoz et al. (2024) show that global economic policy uncertainty (EPU) negatively affects public interest in Bitcoin (measured by Google Trend searches for the keyword Bitcoin), consistent with the view of Bitcoin as a speculative investment rather than a safe haven asset. Khalfaoui et al. (2023) find that in bearish and normal (bull) markets, public attention to the Russia-Ukraine war negatively (positively) affects all cryptocurrencies in the short term. Mohamad (2022) show that the 48 hours around the Russian invasion of Ukraine have led to a flight from the Russian ruble to other foreign currencies (USD and yen) and commodities (silver, brent, and natural gas), and document the presence of herding behaviors between energy commodities and cryptocurrencies (Bitcoin, Ethereum, and Litecoin).

In contrast to traditional safe havens, digital assets, such as cryptocurrencies and stablecoins, depend on blockchain settlement systems, which are essentially networks of peer computers that independently validate transactions and update the transaction ledger according to predefined rules. These peer computers are geographically and jurisdictionally distributed and participate pseudo-anonymously in the network, making it difficult to enforce state-level regulations such as sanctions (Cong and He, 2019, Biais et al., 2023, Nakamoto, 2009). Consistent with this explanation, Alexakis et al. (2024) find a significant increase in global cryptocurrency trading activity

⁵For instance, the 21 million limited supply may not offer enough volume to support the needs of a global reserve currency and introduce deflationary pressure. See also: <https://stacker.news/items/379225>

after the imposition of international sanctions and other financial restrictions on Russia.⁶

Our study differs from these previous studies that have examined the impact of the Russian invasion of Ukraine and the following sanctions on cryptocurrency in a number of ways. First, we establish a clear connection between the timing of the invasion and the follow-up sanctions and the increased interest of *Russian* retail investors in digital assets, particularly in stablecoins, rather than looking at a proxy for public attention to the events as [Khalfaoui et al. \(2023\)](#) do. We then examine the impact of this increased interest, resulting from the events of invasion and sanctions, on *on-chain* metrics rather than on prices and volumes provided by centralized exchanges as in [Alexakis et al. \(2024\)](#). Our analysis of on-chain metrics provides more information on investors' behavior that is not easily captured by the centralized exchange data, which mostly focuses on returns and volume. Moreover, as outlined by [Alexakis et al. \(2024\)](#), a major limitation of relying on centralized exchange data is that exchanges, unlike blockchains, are also prone to sanctions and do not benefit from the blockchain's operation beyond jurisdictional restrictions.

As a result of our novel approach to the question, our findings also contradict some of the findings from this recent literature. Specifically, we contrast [Gill-de Albornoz et al. \(2024\)](#), who find a negative relationship between global political uncertainty and investor attention to Bitcoin, and no impact at all from local political uncertainty. Instead, we find that geopolitical events can affect a specific country and significantly explain the heightened local retail investors' attention and interest in digital assets. We also contradict [Alexakis et al. \(2024\)](#), who find less pronounced effects of Russian-Ukrainian war-related events on stablecoins trading volumes. We instead find that the impact of the invasion and sanctions events on Russian investors' interest and on global on-chain metrics is stronger for stablecoins than for other digital assets. We attribute these differences in findings to our use of on-chain metrics and search interest data, which more accurately capture investor sentiment and behavior, as they are not susceptible to sanctions or financial restrictions.

Our findings contribute to other strands of literature, in addition to the already discussed literature on safe-haven assets' demand as a response to unanticipated events and the safe-haven

⁶Further, some studies show that cryptocurrencies serve as means to facilitate illicit activities and evade sanctions ([Cong et al., 2023](#), [Amiram et al., 2022](#), [Foley et al., 2019](#)).

properties of digital assets. In fact, by demonstrating that some users might be drawn to digital assets as a result of financial sanctions, we also add to the growing literature which examines the characteristics of investors in digital assets,⁷ and the broader literature on the adoption of digital assets (Cong et al., 2021, 2022, Liu et al., 2022, Hackethal et al., 2022, Jin et al., 2023, Murakami and Viswanath-Natraj, 2025).

Furthermore, our paper contributes to the wider literature that explores on-line search interest to gauge changes in retail investors’ attention and their impact on financial markets, and more specifically on the cryptocurrency market, by providing a new methodological tool. In fact, we introduce a novel method for retrieving daily search interest data from Google Trends over arbitrary time periods, addressing a pertinent research limitation in the literature (see, for instance, Barber and Odean (2008), Barber et al. (2009), Barber et al. (2022), and Chava et al. (2024)).

The policy implications of our study are also significant. We demonstrate that digital assets, and stablecoins in particular, that employ settlement systems outside traditional financial regulations and oversight, may offer alternative routes to overcome financial sanctions as they are effectively shielded from such restrictions. This finding should concern entities and countries that have historically relied on the viability of sanctions as a tool for political and economic pressure, and those countries currently facing or likely to face such financial sanctions in the future. The results also contribute to the ongoing debate on the emerging international role of cryptocurrencies, stablecoins, central bank digital currencies (CBDCs) and other digital assets, particularly in relation to their increasing adoption and usage to mitigate financial sanctions. Our work also informs regulators on how to develop regulatory policies to govern these assets, similar to the measures introduced for Initial Coin Offerings (ICO) (Cong and Howell, 2021).

The remainder of the paper is organized as follows. Section 2 discusses our data collection, presents descriptive statistics, and illustrates the methodology. Section 3 details our main empirical results. Section 4 presents the results of all robustness tests. Finally, Section 5 concludes the study

⁷Chava et al. (2024) posit that gambling tendencies predict the attention of retail investors in digital assets. Hackethal et al. (2022) find Bitcoin investors more likely to trade penny stocks. Kogan et al. (2024) compare retail investor positions in stocks and crypto, and other studies show factors such as wealth (Aiello et al., 2022), past returns (Pursiainen and Toczynski, 2022), and COVID-19 stimulus checks (Divakaruni and Zimmerman, 2024) predict future crypto investment.

and provides some further discussion on our main findings.

2. Data and Methodology

2.1. Data and Descriptive Statistics

To proxy the interest of the Russian public in digital assets, we follow [Da et al. \(2011\)](#) and obtain Google search data for keywords related to cryptocurrencies and stablecoins from Google Trends. [Chava et al. \(2024\)](#) also use Google Trends to gauge the public’s interest in cryptocurrencies. We selected Google Trends over other sources of data proxying public interest, such as X (formerly Twitter), LexisNexis, or Yandex Worldstat (a product similar to Google Trends that is related to Yandex, a popular search engine in Russia), for a number of reasons. First, Google is the world’s most popular search engine while X has recently limited their API access,⁸ rendering it impractical for search interest analysis. Second, LexisNexis, being subscription-based and subject to retroactive removal or addition of news items, introduces potential bias and lacks transparency in its data coverage. Third, Google Trends has been shown to reflect the profiles of more diverse investors and is better suited to address country- and area-specific interest ([Jun et al., 2018](#), [Chava et al., 2024](#)). Fourth, Yandex Worldstat only provides daily data for the last 60 days. Access to weekly or monthly data is also limited to short time periods (e.g., three consecutive months at a monthly frequency).

To determine whether there is a substantial difference between Google Trends and Yandex Worldstat, we follow [Dimpfl and Jank \(2016\)](#) and compare the search interest for keywords in each search engine by calculating the correlations. We find high correlations for unambiguous keywords⁹ related to stablecoins and cryptocurrencies, such as correlations over 0.9 for search interests in ‘*USDT*’ and ‘*ETH*’ across the two search engines, and over 0.7 for ‘*BTC*’. For terms subject to spelling variations, we find correlations of 0.6 for ‘*Binance*’ and ‘*USD*’. This suggests that search trends align well between the two platforms, even for terms with minor spelling differences. This

⁸An API, or application programming interface, is a set of rules or protocols that enables software applications to communicate with each other to exchange data, features and functionality. For more information, see: <https://x.com/XDevelopers/status/1641222782594990080>

⁹Ambiguous terms are those with variations in spelling, such as *Binance*, *Бинанс*, or *Binanse*.

means that we should be largely indifferent to using Google Trends or Yandex as our data source. We opt to use Google Trends, noting also that Russia’s internet restrictions may subject Yandex data to higher scrutiny compared to Google Trends.

Next, we perform a correlation test to determine whether utilizing a single keyword or multiple related keywords combined into an index would be a better measure of public attention. We find that using a single keyword is sufficient to capture attention. For instance, in Google Trends we observe correlations above 0.95 between the Latin and Cyrillic alphabet versions of terms like *dollar* and *USD* or *Binance* and *Бинанс*, as well as correlations over 0.9 between specific cryptocurrencies’ extended names and their ticker symbols (e.g., *Bitcoin* and *BTC*, *Ethereum* and *ETH*, and *Tether* and *USDT*). We select the most popular keywords related to major categories of the cryptocurrency ecosystem: for stablecoins we use *USDT* as keyword, for cryptocurrencies that have a role as a store of value we use *Bitcoin*, for utility cryptocurrencies used for gas and smart contract execution and underpinning decentralised applications we use *Ethereum*, for centralised exchanges we select *Binance*, for decentralised exchanges *Uniswap*, and for self-custody solutions we use *Metamask*. We reasonably assume that by picking the most popular asset/entity in each category, these keywords may also be familiar to non-sophisticated retail investors, whose behavior may be revealed by search interest trends on the Google search engine. We query each of these keywords on Google Trends. If the daily search interest is zero for more than a month, then the keyword (and its respective category) is dropped, as we deem it may be unpopular with retail investors. Subsequently, we end up using the following keywords: *USDT*, *BTC*, *ETH*, *Binance*, and *USD*, which are summarized along with justifications in Table 1. For each of these keywords, we source search interest data from the start of 2021 to the end of 2023 and for different regions (Russia, Ukraine, the US, and globally). Google Trends normalizes the *daily* search interest based on the time and location of each query. To be more specific, Google Trends captures a term’s relative popularity by comparing its search activity on a particular date within a specific geographic region and time frame. The resulting metric is scaled over a range going from 0 to 100 and is called the Google Search Volume Index (SVI).

The studies we have reviewed that utilize search engine data from Google Trends typically ex-

amine longer time periods than ours, for which weekly data frequency is appropriate. As explained in the next section 2.2, we adopt an event study methodology that requires a measure of search engine interest for selected keywords at daily frequency, which Google Trends does not provide for periods longer than 90 days. Hence, we develop an original methodology to obtain daily data from Google Trends. We first collect daily Google Trends data for each keyword over sub-periods of 90 days, progressively. Simply merging these data would result in an inaccurate time series since Google Trends provides scaled data relative to the highest and lowest points within each specified 90-day period. Consequently, merging consecutive 90-day windows of data downloaded from Google Trends would not yield a time series of comparable observations. To avoid this challenge, we employ a standardization technique that involves overlapping consecutive 90-day period queries by one day. This overlap allows us to directly compare the values at these dates and identify a scaling factor. Specifically, given two 90-day windows of data w_1 and w_2 , overlapping by one day, our standardization process requires calculating a factor f :

$$f = \frac{V_{w_2,1}}{V_{w_1,90}} \quad (1)$$

where $V_{w_1,90}$ is the last observed value in Google Trends for a certain keyword in the 90-day window w_1 (hence it is the observation that overlaps with the next consecutive rolling period of 90 days w_2) and $V_{w_2,1}$ is the first observed value in Google Trends for the same keyword in the next 90-day window w_2 .

Using this factor f , we then adjust every observation n in the next 90-day period w_2 to align the scales:

$$V_{w_2,n}^{\text{adj}} = \frac{V_{w_2,n}}{f} \quad (2)$$

This approach has a limitation that arises when $V_{w_1,90}$, the overlapping value for our specific keyword, is zero, preventing scaling. To resolve this limitation, when the overlapping value is found to be zero, we adjust the starting point of window w_2 so that its first day corresponds to the date of the last non-zero value $V_{w_1,k}$ in window w_1 . We then recompute the scaling factor f

using this value:

$$f = \frac{V_{w_2,1}}{V_{w_1,k}} \quad (3)$$

Here, k denotes the day in window w_1 where the last non-zero value $V_{w_1,k}$ is found (with $k < 90$), and $V_{w_2,1}$ is now the value in w_2 that corresponds to the same calendar date as $V_{w_1,k}$. If no non-zero value is found on any day over the 90-day period w_1 , we proceed by not scaling observations in the next 90-day periods. The scaling process is described below.

$$f = \begin{cases} \frac{V_{w_2,1}}{V_{w_1,90}}, & \text{if } V_{w_1,90} \neq 0 \\ \frac{V_{w_2,1}}{V_{w_1,k}}, & \text{if } V_{w_1,90} = 0 \text{ and a non-zero value } V_{w_1,k} \text{ is found} \\ 1, & \text{otherwise.} \end{cases} \quad (4)$$

In addition to Google Trends data for public interest around some keywords, we also source a variety of on-chain data from Coin Metrics, related to the cryptocurrencies used as our Google Trends keywords. The metrics include addresses activity, holding amounts, transactions, and native unit amounts.

In contrast to previous literature on the impact of the Russian invasion on cryptocurrencies (e.g., [Alexakis et al. \(2024\)](#)), we are the first to use on-chain data instead of exchange data and not rely exclusively on returns metrics. This is important for two reasons: first, it provides additional information on cryptocurrency adoption and investor behavior that is not captured by price data. Secondly, we do not rely on Centralized Exchange (CEX) volumes, which can be affected by sanctions because they do not benefit from the pseudo-anonymity and geographical distribution of blockchains.

Next, we obtain data on the sanctions imposed by the EU and the US on Russia from the Open Sanctions database.¹⁰ An indicative list of the sanctions data collected can be found in Table 2. Finally, we collect data for a range of financial variables: S&P500 adjusted returns, crude oil futures prices, gold futures prices, exchange rate RUB to USD, and CBOE volatility index as

¹⁰<https://www.opensanctions.org/>

control variables in our analysis.

Tables 3 presents the descriptive statistics for all variables in our study. Panel A shows that the stablecoin keyword *USDT* shows a higher average search interest in Russia and Ukraine than in the US and globally. In Panel B, we observe that USDT prices remain stable (deviations from fiat USD value average to zero). Across different blockchains (Ethereum, Tron), stablecoins are predominantly held by small to medium holders (i.e. active addresses with lower values), but larger transactions dominate. Similar trends are observed for Bitcoin and Ether.

2.2. Methodology

To investigate the existence of a link between the events of the invasion of Ukraine, the subsequent imposition of sanctions on Russia, and changes in public attention and hence retail investors' interest in stablecoins and cryptocurrencies, we employ a differences-in-differences (DiD) approach, where we compare the keywords search interest in Russia against global (WO) interest around the key events. For robustness, we also compare the retail investors' interest in Russia against the United States (US) and then against Ukraine (UA).

Ukraine was selected as a control group due to its direct participation in the conflict but lack of exposure to sanctions, allowing us to isolate the effects of sanctions from those related to the war itself. The US serves as a reference point because of its status as a significant market for digital assets. We also use the world (WO) as a further control group to identify similar or differential trends across all regions. Given the nature of search engine data, the WO control group includes search data from Russia too; however, we expect this not to have any meaningful impact, as Russia constitutes only a small subset of the global data. Also, if any impact does occur, this can only bias the results against our hypothesis (i.e., against demonstrating any differential effect of the war and sanctions in Russia versus the control group).

Before implementing our DiD model, we demonstrate that our 'treated group', i.e., Russia, and the 'control groups', i.e., WO, US and UA exhibit parallel trends before the invasion for all

keywords in our sample. Our parallel trends test model is specified as follows:

$$\begin{aligned}
\mathbf{Y}_{it} = & \beta_0 + \beta_1 \text{Russia}_i + \beta_2 \text{Jan2022}_t + \beta_3 \text{Nov2021}_t \\
& + \beta_4 \text{Dec2021}_t + \beta_5 \text{Russia}_i \times \text{Feb2022}_t \\
& + \beta_6 \text{Russia}_i \times \text{Nov2021}_t + \beta_7 \text{Russia}_i \times \text{Dec2021}_t + \epsilon_{it}
\end{aligned} \tag{5}$$

where $\mathbf{Y}_{it}$ is a vector representing the Google Trends values for search interest (and retail investors' attention) for each selected keyword (*USDT*, *BTC*, *ETH*, *Binance*, *USD*) in region i at time t . β_0 is the constant or intercept that captures the baseline level of search interest when all other variables are equal to zero. Russia_i is a dummy variable that equals 1 if the observation is related to Russia and 0 if it is related to any of the control groups (the world, the US, or Ukraine). Jan2022_t , Nov2021_t , and Dec2021_t are time dummies which equal one if the observations are for the months of January 2022, November 2021, and December 2021, respectively; and zero otherwise. These dummies are used to capture general changes in search interest that have equally affected all regions during these months. The interaction terms ($\text{Russia}_i \times \text{Feb2022}_t$), ($\text{Russia}_i \times \text{Nov2021}_t$), and ($\text{Russia}_i \times \text{Dec2021}_t$) represent the differential effect of these time periods in Russia compared to the world (or the US or Ukraine), indicating whether there is a significant change in search interest specific to Russia during these months.

Next, we employ two DiD models. The first examines the short-term effects of the events of the invasion of Ukraine and the subsequent imposition of sanctions on Russia on stablecoin and cryptocurrency search interest focussing on six months after the invasion. The second captures the longer-term effects of the events, extending the analysis to the end of 2023.

The generic DiD model is specified below:

$$\begin{aligned}
\mathbf{Y}_{it} = & \beta_0 + \beta_1 \text{Russia}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Russia}_i \times \text{Post}_t) \\
& + \beta_4 \mathbf{Controls}_t + \gamma_t + \epsilon_{it}
\end{aligned} \tag{6}$$

where $\mathbf{Y}_{it}$ is a vector representing the Google Trends values for search interest (and retail investors' attention) for each selected keyword (*USDT*, *BTC*, *ETH*, *Binance*, *USD*) in region i at time t .

β_0 is the constant or intercept, which captures the baseline level of search interest when all other variables are equal to zero. Russia_i is a dummy variable that equals 1 if the search interest observation is related to Russia and zero otherwise (if related to the World, the US or Ukraine). Post_t is a dummy variable that equals 1 for the post-invasion period (six months after February 24, 2022 for the model looking at short-term effects, and from February 24, 2022 until the end of 2023 for the model looking at longer-term effects), and 0 otherwise. ***Controls_t*** represents a vector of control variables (S&P500 adjusted returns, crude oil futures prices, gold futures prices, RUB to USD exchange rate, and CBOE Volatility Index), γ_t denotes year (i.e., 2022, 2023, etc.) and month fixed effects (i.e., January, February, etc.), and ϵ_{it} is the error term.

We further employ a dynamic DiD approach to examine how the effect of the invasion and the ensuing sanctions on retail investors' interest in Russia evolves over time. Specifically, we focus on how the interest of retail investors evolves from the initial shock (initial invasion in February 2022) to the following year. We estimate the following model:

$$\mathbf{Y}_{it} = \alpha + \sum_{j=1}^2 \beta_j \text{Year}_{j,it} + \gamma \text{Russia}_i + \sum_{j=1}^2 \delta_j (\text{Year}_{j,it} \times \text{Russia}_i) + \boldsymbol{\theta} \mathbf{Controls}_t + \epsilon_{it} \quad (7)$$

where $\mathbf{Y}_{it}$ represents the vector of search interest values for our selected keywords for country i in time period t . Year_j represents a time dummy for the respective years (2022 and 2023, indexed by j), while Russia_i is the indicator variable for Russia. The interaction terms $\text{Russia} \times \text{Year}_{j,it}$ capture post-invasion differences in Google Trends search interest between Russia and the world.

Finally, for robustness, we employ a Regression Discontinuity in Time (RDiT) approach. More specifically, the RDiT model is applied to retail investors' interest in Russia for each of the selected keywords and for a period spanning from six months before the invasion (February 2022) to six months after. Our baseline RDiT model is specified as follows:

$$\mathbf{Y}_t = \beta_0 + \beta_1 \text{Post}_t + \beta_2 \text{Day}_t + \beta_3 (\text{Post}_t \times \text{Day}_t) + \boldsymbol{\beta}_4 \mathbf{Controls}_t + \epsilon_{it} \quad (8)$$

where $\mathbf{Y}_t$ represents now the vector of search interest values for the selected keywords in our

sample only for Russia at time t . The term Post_t is a binary variable that takes the value of one if the observation is from the post-invasion period (i.e., after February 24, 2022), and zero otherwise. The term Day_t represents the number of days relative to the invasion date, where positive values indicate days after the invasion and negative values represent days before the invasion. The interaction term ($\text{Post}_t \times \text{Day}_t$) captures any differential impact of the invasion of Ukraine on the time trends of the search interest from Russians for specific keywords related to stablecoin and cryptocurrencies in the post-invasion period. The vector ***Controls_t*** represents the same set of control variables used in Equation (6).

3. Empirical Analysis and Results

3.1. Impact of invasion and sanctions on retail investors' interest in Russia

Table 4 summarizes the results of the parallel trend analysis between search interests in Russia and worldwide for the five selected keywords (*USDT*, *BTC*, *ETH*, *Binance*, *USD*). We also plot the search interest for *USDT* in Russia and globally in Figure 1. The parallel trends tests between Russia and the US, as well as between Russia and Ukraine are reported in the Appendix at Tables A1 and A2. Overall, we observe that the retail investors' interest for the selected keywords satisfy the parallel trends assumption, as in the periods of no 'treatment' (before the invasion and the subsequent sanctions, that is, from December 2021 to January 2022) the difference in search interest between the 'treated' group (Russia) and the 'control' groups (world, the US, and Ukraine) is insignificant. In fact, as Table 4 shows, except for one instance (Ethereum in December 2021, where the interaction term $\text{Russia} \times \text{Dec2021}$ is significant at the 10% level), Russian retail investors' interest for any of the keywords used does not differentiate from the worldwide interest.¹¹ Therefore, we can conclude that our data satisfy the parallel trends assumption required for DiD.

Table 5 presents the short-term DiD results (where the dummy 'Post' is equal to one over the six months following the invasion) comparing Russia with the world, and Table 6 extends 'Post'

¹¹Similarly, the parallel trends assumption is satisfied for the keywords *USDT*, *BTC*, and *Binance* between retail investors' interest in Russia and Ukraine, and for the keywords *BTC* and *USD* between retail investors' interest in Russia and the US. Even in those few instances where disruptions to the parallel trends assumption are observed, they appear temporary and not for the entire observed period.

to the end of 2023. The DiD tests' results obtained using Russia as the treated group and the US or Ukraine as control groups are presented in the Appendix at Tables A3 and A4, respectively.

Taken together, the findings in Tables 5 and 6 indicate that even after controlling for additional factors and using year- and month-fixed effects, in Russia there is a significant increase in retail investors' interest in digital currencies following the invasion date and during the period of imposition of sanctions. The increase in interest and attention is particularly strong for *USDT*.

Specifically, when comparing the interest of Russian retail investors with the global one (see Table 5, last column), we observe that the search interest in *USDT* is initially 2% lower in Russia than elsewhere in the world, as indicated by the negative coefficient of the dummy variable *Russia*. Following the invasion, we find a significant 5% increase in search interest in all countries, as evidenced by the positive coefficient of the *Post* dummy variable. Most importantly, the coefficient of the interaction term $Russia \times Post$ is positive, statistically significant, and economically important (16% increase). Hence, our results reveal that while interest in stablecoins was initially lower in Russia than globally, the invasion of Ukraine has led to a 16% increase in search interest in Russia above the global average. This represents the largest increase in search interest observed across all of our selected keywords. The coefficient of the interaction term $Russia \times Post$ shows the highest positive value for *USDT* followed by *Binance*, while for *ETH* and *BTC* there is a positive impact, but it is lower in magnitude, and for *BTC* it is not statistically significant. We find statistically significant increases for interest in *ETH* and *Binance*, which also exhibit similar trends: after the invasion, there is an 8% increase in Russian retail investors' interest for *ETH* and a 15% increase for *Binance*, despite the global decline in search interest for the two keywords. Finally, during the period of economic sanctions, we also observe a 12% higher search interest for *USD* in Russia than globally.

Table 6 shows similar results for retail investors' search interest in cryptocurrencies, albeit slightly diminished in magnitude.¹² Similar effects are also observed when comparing Russian retail investors' interest in the selected keywords to other control groups. The last column of

¹²The search interest in *USD* in Russia shows instead a 6% decrease compared to the global average in the period after the invasion of Ukraine.

Appendix Table A3 shows that retail investors’ interest in *USDT*, *BTC*, *ETH*, *Binance*, and *USD* increased by 20%, 5%, 11%, 5%, and 15% respectively more in Russia than in the US during the six months following the invasion. Furthermore, Appendix Table A4 reveals that retail investors’ interest in *USDT*, *BTC*, and *USD* rose by 4%, 5%, and 15%, respectively, more in Russia than in Ukraine. Finally, Appendix Tables A5 and A6 extend these DiD analyses to the entire post-invasion sample period, yielding overall similar results for keywords related to cryptocurrencies.

Our dynamic DiD analysis provides further insight into how these effects evolve over time. Table 7 compares Russia with the world, demonstrating that the post-invasion differential increase in Russian retail investors’ interest for digital asset keywords — *USDT*, *BTC*, and *Binance* — observed in 2022 not only persists, but also strengthens in 2023. For instance, the interest in *USDT* rises by 8% in 2022 and increases further to 9% in 2023. Similarly, the interest in *Binance* grows by 11% in 2022 and reaches 13% in 2023. In contrast, *BTC* interest shows a comparatively modest increase, rising by a statistically significant 3% in 2023.

Overall, we find that the initial surge in retail investors’ interest in digital assets in Russia not only persists but, in some cases, intensifies in the year following the invasion and the imposition of sanctions. These effects are particularly notable for the keywords *USDT* and *Binance*, possibly suggesting the increased attention of retail investors to digital alternatives that could mitigate the exclusion from international financial markets and payment systems due to the imposed sanctions.

3.2. Impact of Russian retail investors’ interest on digital asset adoption metrics

Next, we examine the impact of changes in Russian retail investors’ interest on various on-chain metrics for the digital assets in our sample. Specifically, we now use the changes in interest of retail investors captured from Google Trends as our independent variable and the changes in on-chain metrics as our dependent variables. The time period of this analysis goes from February 2022 until the end of 2023 to account for the effects of the invasion and the following sanctions.

First, we analyze how changes in search interest in the keyword *USDT* relate to changes in *USDT* on-chain metrics for the two major blockchains where *USDT* is issued: Ethereum and Tron (see Tables 8 and 9, respectively). Next, we examine the relationship between changes in search interest for the keywords *BTC* and *ETH* and changes in Bitcoin and Ether on-chain metrics (the

results are illustrated in Tables 10 and 11, respectively). In addition, we look at how changes in search interest for the keyword Binance impact changes in Bitcoin and Ether flows to and from Binance (see Table 12). Finally, we use changes in USD search interest as the independent variable, with the dependent variables being changes in USDT on-chain metrics (Tables 13 and 14), since USDT and USD can be considered close substitutes. In all these regressions, we control for year and month fixed effects as well as for various control variables (daily changes in: RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices).

We believe that our reliance on on-chain metrics overcomes two key limitations in the existing literature. First, on-chain data can provide more insights into investors' behavior that is not captured by centralized exchange data, which often just focus on returns. Secondly, exchanges are subject to sanctions and other regulatory constraints, while blockchains operate beyond such restrictions.

Our results suggest that the increased search interest in the keywords USDT and USD is not significantly associated with price fluctuations for USDT on either Ethereum or Tron (see Tables 8, 9 13, and 14). This indicates that if this increased interest translates into higher demand over the period analyzed, the stablecoin is able to absorb the additional demand without significant deviations from its peg.

Table 8 reveals that the percentage of USDT available supply held in addresses with balances between \$1,000 and \$10,000 US dollars exhibits a statistically significant increase in response to a heightened search interest for the keyword USDT. Similarly, we observe a statistically significant increase in the percentage of USDT available supply held in addresses with balances between \$100 and \$1,000 US dollars on the Tron blockchain (Table 9). Hence, the addresses impacted by the increase in search interest in USDT are those of relatively smaller size (compared to the others we have examined for which we find no statistically significant change in holdings). This result suggests that an increase in USDT online search interest in the sanctioned economy (Russia) is associated with an increase in the stablecoin holdings by retail users, who are typically those with smaller USDT balances in addresses. It is also sensible to consider that smaller holdings on the two blockchains (Ethereum and Tron) are identified by different ranges of USDT balances in addresses.

Ethereum has higher fees compared to Tron, which can even reach a few dollars per transaction. Therefore, it can be rather uneconomical to transact in very small amounts on Ethereum. Retail investors' addresses on Ethereum may then hold relatively higher balances. Tron, on the other hand, has lower fees and can accommodate smaller transactions, and hence even smaller holdings by retail investors.

Moreover, in Tables 8 and 9 we observe that as a result of the increase in interest in USDT, the volumes of transactions for the stablecoin increase in both networks, but only for larger transactions. Instead, there is no significant increase in volumes for transactions of smaller amounts. This suggests that the increase in interest in the keyword USDT does not prompt trading from retail investors, who hold smaller amounts and can only engage in transact of smaller size. Conversely, retail investors seem to tend to accumulate more USDT holdings and use this stablecoin as a digital safe haven.

For Bitcoin, Table 10 shows that increased search interest in the keyword BTC is associated with an increase in BTC returns. Additionally, there is an increase in active, receiver, and sender addresses as well as in transaction volumes, particularly for larger amounts of BTC. For Ether on-chain metrics (Table 11) the analysis shows that an increase in search interest for the keyword ETH is associated with a higher amount of ETH in addresses holding larger balances.

Lastly, we observe that a heightened search interest for the keyword Binance is associated with a statistically significant increase in outflows of both BTC and ETH from Binance (but no significant change in inflows). This may suggest that users may be cautious of the effects of sanctions on entities that operate within the traditional financial system (such cryptocurrency exchanges which interface with banks and KYC/AML providers) and consequently may withdraw their assets from Binance to avoid having them frozen or seized.

Overall, our findings suggest that the heightened search interest in the stablecoin keyword *USDT* in Russia during the post-invasion period is associated with an increase in stablecoin holdings in smaller addresses that are typically for retail investors. However, it is not associated with an increase in smaller-size transaction volumes, which suggests the prevalence of stablecoin accumulation from retail investors and therefore a tendency toward using the stablecoin as a safe

haven. Our results suggest that for Russian investors in the post-invasion and sanctions period, the US dollar-backed stablecoin has become a substitute for international currency, and digital assets have allowed Russian capital to be invested globally and to escape the sanctions limitation.

4. Robustness Tests

To ensure that the observed increase in search and Russian public interest in cryptocurrencies, and USDT in particular, after the invasion of Ukraine and during the sanctions’ period is not driven by random fluctuations or unrelated factors, we conduct a series of placebo DiD tests by shifting the treatment period to six months before the invasion (August 2021) and limiting the post-treatment period to the months prior to the invasion. These placebo tests allow us to examine whether similar patterns emerge in a period when no sanctions are imposed on Russia as a result of the invasion. Table 15 summarizes our results. In all placebo tests, the coefficient of the interaction term $Russia \times Post$ is not positive, suggesting that the increase in the interest of Russian retail investors is specifically associated with the post-invasion period and the resulting sanctions. In fact, for the keywords *USDT*, *ETH*, and *USD*, the estimated coefficients of the interaction terms are negative, suggesting that Russian search interest in certain categories of digital assets was below the global average level in the period before invasion (i.e., the treatment period of this placebo test). $Russia \times Post$ is statistically insignificant for *BTC* and *Binance*, suggesting that the increased search interest for these keywords is specific to the invasion period and does not arise in an arbitrary pre-invasion period. This reinforces the robustness of our findings, confirming that the changes in Russian retail investors’ interest in cryptocurrencies are linked to the invasion event.

In addition to using a DiD approach, we also conduct a regression discontinuity analysis (RDiT) to provide additional evidence of the association between the invasion event and sanctions and an increase in Russian retail investors’ interest in digital assets. Unlike our previous DiD analysis, RDiT relies exclusively on the Russian sample of observations to detect significant changes in search interest. The analysis focuses on the window that spans six months before and after February 2022. As summarized in Table 16, the RDiT-estimated coefficient of *Post* indicates that for *USDT* there is a 29% increase in retail investors’ interest after the invasion, while the search interest in *USD*

risers by 31%. Similarly, the search interest in *Binance* increases by 10%. For *USDT*, *USD*, and *Binance* the effect is statistically significant at the 1% level. The negative coefficients of the interaction terms *Russia* $\times$ *Days* suggest that although the search interest in the five keywords spikes immediately after the invasion, it gradually decreases as the post-invasion period progresses.

We then implement a Donut RDiT approach by excluding the observations in the immediate months around the cut-off date (February 2022). By removing these data points, we ensure that the detected discontinuity reflects the true treatment effect, not contaminated by the influence of unrelated observations near the cutoff date. Table 17 shows that the Donut RDiT analysis confirms our previous findings in Table 16. For instance, we observe a significant post-invasion increase in search interest for *USDT* (28%) and *USD* (22%), which are both highly statistically significant at the 1% level. We observe similar but less pronounced results for *BTC*, *ETH*, and *Binance* which show an increase of 5%, 6%, and 3%, respectively, confirming that stablecoins such as USDT can capture more public attention in response to the invasion.

Finally, we study the impact of individual sanctions on measures of ‘cumulative abnormal retail investors’ interest’(CAI) specified as follows:

$$CAI_t = \sum_{t_0}^t (\text{Search Interest}_t - \text{Expected Interest}_t) \quad (9)$$

where Search Interest_t is the observed retail investors’ interest in a specific keyword related to a digital asset or infrastructure, and $\text{Expected Interest}_t$ is its expected value estimated from the pre-event period. We use a pre-event estimation window going from 90 to 10 days before the event, while the event window covers 10 days before to 10 days after the event.

Overall, the findings summarized in Table 18 show that, when statistically significant, sanctions have a positive impact on the cumulative abnormal search interest, with a stronger effect on the keywords *USDT*, *USD*, and *Binance*. Indicatively, the search interest in *USDT* and *USD* shows sharp spikes immediately following key sanction dates in February, March, and April 2022. These results are significant at the 1% level, with similar patterns observed in the other selected keywords. In addition, we observe what we term the ‘anniversary effect’, where statistically significant spikes in public retail investors’ interest resurge as a consequence of sanctions one year

after the invasion. This effect likely arises from the resurfaced public awareness and media coverage surrounding the invasion and the sanctions, particularly around key dates marking the start of the conflict. During these periods, concerns about the lasting economic consequences of the sanctions and the geopolitical situation re-emerge, resulting in renewed spikes in retail investors' interest. Furthermore, this effect may be tied to the fact that on these anniversaries, the parties that imposed sanctions often reaffirm their commitment to maintaining or even tightening them. These reaffirmations serve as reminders of the ongoing restrictions, prompting renewed interest in alternative assets such as digital currencies.

5. Discussion and Conclusions

Our study examines the heightened interest and engagement of retail investors in digital assets in Russia following the 2022 invasion of Ukraine and the subsequent imposition of financial sanctions. Using online search interest data, we investigate the effects the invasion and sanctions had on Russian retail investors' attention towards digital asset-related keywords. We then assess whether this heightened interest translates into actual behavior by analyzing the reaction of on-chain metrics to retail investors' activity.

Our study is novel in several respects. It is the first to explore investors' attention towards digital assets as a consequence of war events. We are also the first to investigate how this increased interest influences changes in on-chain metrics. In this context, we introduce a novel methodology for extracting *daily* search interest data from Google Trends. Furthermore, our use of on-chain data on prices, transactions, holdings, and addresses activities offers unique insights into investors' behavior that are not captured by centralized exchange data, which predominantly focuses on returns. Also, importantly, centralized exchanges are subject to sanctions and regulatory constraints, whereas blockchains operate independently of such restrictions.

The significance of our novel approach lies in its ability to challenge the existing literature in the field. For instance, previous studies have found less pronounced effects of the Russian-Ukrainian war on stablecoins; however, our findings diverge due to our use of on-chain data and daily search interest metrics, which we argue provide a more accurate representation of retail investors' sentiment and behavior.

Overall, we find that, in response to the invasion and sanctions, in Russia there is a significant surge in retail investors’ interest in the USD-denominated stablecoin USDT, as well as cryptocurrencies like Bitcoin (BTC) and Ether (ETH), and platforms such as Binance. This elevated interest not only persists, but, in some cases, intensifies over the subsequent year, indicating that events such as sanctions trigger a sustained shift in public interest towards digital assets. We also identify “anniversary effects”, where spikes in the interest of retail investors occur a year after the invasion started. These spikes likely stem from the resurgence of public awareness and media coverage surrounding the invasion and the sanctions, particularly on key dates marking the start of the conflict.

Importantly, this increased attention of retail investors correlates with observable changes in on-chain metrics. Specifically, we observe an increase in USDT holdings in smaller addresses, increased on-chain USDT transactions, and a tendency toward USDT accumulation rather than short-term trading. These patterns suggest that the stablecoin is perceived as a digital safe haven during periods of geopolitical and financial instability. Furthermore, we observe users withdrawing their cryptocurrency holdings from exchanges, which we attribute to fear over the effect sanctions on regulated cryptocurrency intermediaries.

Moreover, despite the heightened interest that fuels the increase in demand, USDT successfully maintains its peg to the USD, demonstrating its robustness in absorbing increased usage and its dependability during periods of financial uncertainty. This reliability could encourage similar shifts toward stablecoins in future geopolitical or economic crises.

Our results indicate that the Russian public views digital assets, particularly the USD-denominated stablecoin USDT, as viable alternatives to safeguard wealth and mitigate the effects of financial sanctions. The reliance on blockchain settlement mechanisms, which operate independently of traditional financial systems and are less susceptible to state-level interventions, makes these assets especially appealing during periods of economic uncertainty and geopolitical tension. This trend is concentrated among smaller holders on Tron blockchain, signalling increased adoption among retail investors rather than whales or institutional participants.

Blockchains also offer pseudo-anonymity for smaller holders, as tracking and linking them to

real-world identities would be economically inefficient. In contrast, if large institutional players were to use blockchains to evade sanctions on a larger scale, sufficient resources could be deployed to reveal their identity. This creates a differentiated effect, where small holders benefit disproportionately compared to larger holders or institutional investors due to the high costs associated with identifying them. As a result, blockchains contribute to “democratizing” access to the financial system during periods of financial restrictions, particularly benefiting smaller and comparatively underprivileged holders.

Our findings have significant implications for policymakers and regulators, particularly in the context of other existing geopolitical conflicts. The ability of individuals to bypass traditional financial systems using blockchain-based assets undermines the effectiveness of sanctions as a tool for economic pressure. To maintain their efficacy, strategies based on sanctions must be re-evaluated to address the decentralized nature of digital assets. This will likely require international cooperation to develop regulatory frameworks capable of effectively monitoring and, where necessary, restricting financial activities that can bypass sanctions, without stifling innovation in the digital asset sector.

Furthermore, the accelerated adoption of stablecoins in response to sanctions could influence the approaches of central banks to developing and implementing CBDCs. This is particularly relevant for wholesale CBDCs, as they can provide a “state-sponsored” mechanism for cross-border capital flows, helping central banks maintain monetary sovereignty in a rapidly evolving financial landscape.

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Figures and Tables

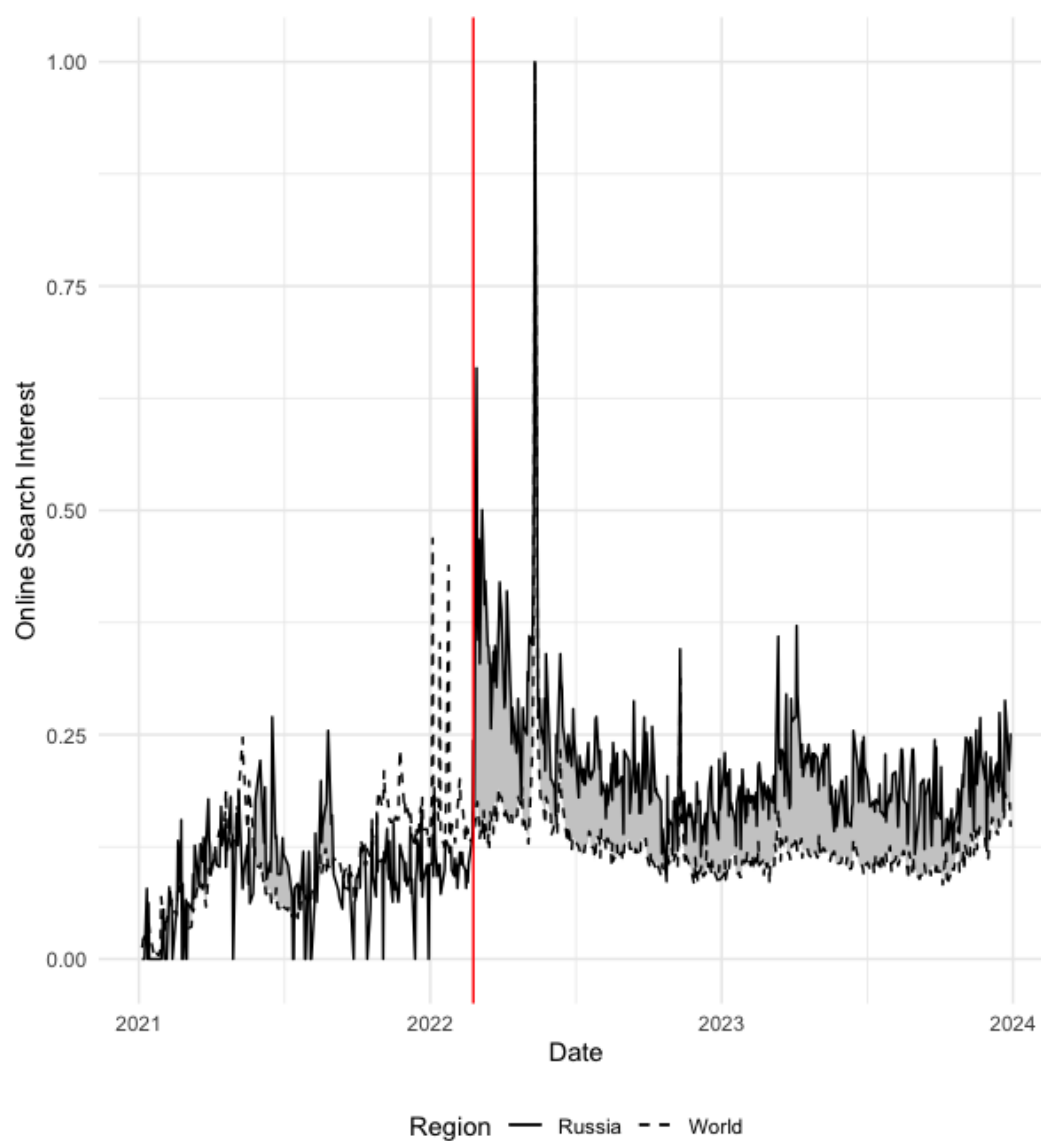


Figure 1: **Online search interest in Russia and the world.**

Online search interest in Russia and the world over time. The event of Russian invasion of Ukraine is represented by the red vertical line. The grey-shaded area represents the excess retail investor interest in Russia versus the world.

Table 1: Selected Keywords for the Cryptocurrency Ecosystem and Justifications.

This table presents the selection of popular keywords representing key components of the cryptocurrency ecosystem. Column “Keywords” lists the specific terms chosen for analysis. Column “Justification” explains why each keyword was selected, based on its relevance to retail investors and its role in the cryptocurrency market.

Keywords	Justification
USDT	Stablecoin, popular as part of the cryptocurrency ecosystem and familiar to retail investors.
BTC	Bitcoin, widely known as the most popular store of value in the cryptocurrency ecosystem.
ETH	Ethereum, the leading cryptocurrency for smart contracts and decentralized applications.
Binance	Binance, the most popular centralized exchange within the cryptocurrency ecosystem.
USD	The US dollar, included for its role in stablecoins and the broader cryptocurrency ecosystem.

Table 2: Indicative List of Sanctions.

This table presents an excerpt of sanction events collected from Open Sanctions. Column “Date” reports the date that the sanction was announced. Column “Total” reports the total number of sanctions. Columns “Companies”, “Individuals”, and “Other” report the number of companies, individuals, or other entities targeted by the sanction. Column “Entity” reports the sanctioning entity, with only sanctions from the EU and the US included in our data. Column “Description” provides a brief description of the sanction, based on the authors’ assessment. The column “Description” is not utilised in our statistical analysis.

Date	Total	Companies	Individuals	Other	Entity	Description
21/04/2022	1	0	1	0	EU	Asset freezes on oligarchs.
08/04/2022	88	72	1	15	US	Expansion of trade sanctions on Russian goods.
08/04/2022	23	0	23	0	EU	Ban on supply of military equipment.
15/03/2022	3	0	0	3	US	Restrictions on imports of Russian natural gas.
15/03/2022	6	0	6	0	EU	Expansion of trade sanctions.
14/03/2022	1	0	0	1	EU	Restrictions on energy imports.
11/03/2022	6	0	6	0	EU	Exclusion of Russian banks from SWIFT.
10/03/2022	81	73	3	5	US	Travel bans on Russian oligarchs.
09/03/2022	3	0	3	0	EU	Travel bans on government officials.
04/03/2022	48	43	0	5	US	Exclusion of Russian banks from the SWIFT system.
28/02/2022	6	0	6	0	EU	Asset freezes on major banks.
25/02/2022	32	32	0	0	US	Initial financial sanctions following the invasion of Ukraine.
25/02/2022	74	0	74	0	EU	Initial trade sanctions following the invasion of Ukraine.
23/02/2022	3	2	0	1	US	Sanctions on individuals involved in recognizing Donetsk and Luhansk.
23/02/2022	6	0	6	0	EU	Sanctions on individuals involved in recognizing Donetsk and Luhansk.
21/02/2022	2	0	2	0	EU	Restrictions on Russian sovereign debt.

Table 3: Panel A: Descriptive statistics for retail investor interest and control variables.

This table reports descriptive statistics for retail investor interest in USD-backed stablecoin Tether (USDT), Bitcoin (BTC), Ethereum (ETH), Binance, and USD across Russia, Ukraine, the United States, and the world (WO) during the period of interest. Additionally, the table reports descriptive statistics for daily changes in the control variables: RUB/USD exchange rates, the S&P 500 index, gold price, oil price, and the VIX index. The statistics presented include the mean, standard deviation (sd), median, minimum (min), maximum (max), range, skewness, kurtosis, and standard error (se). Each variable has 1,095 observations. The retail investor online search interest values for each keyword range between 0 and 1. Changes in control variables are expressed in their respective units (RUB/USD exchange rate, S&P 500 index points, USD per ounce for gold, USD per barrel for oil, and index points for the VIX).

	mean	sd	median	min	max	range	skew	kurtosis	se
Online Search Interest									
USDT in Russia	0.00	0.06	0.00	-0.71	0.49	1.20	-1.16	35.30	0.00
BTC in Russia	-0.00	0.05	-0.00	-0.52	0.56	1.07	1.29	40.10	0.00
ETH in Russia	-0.00	0.07	-0.00	-0.70	0.49	1.19	-0.86	22.96	0.00
Binance in Russia	0.00	0.05	0.00	-0.74	0.48	1.22	-3.22	79.87	0.00
USD in Russia	0.00	0.07	0.00	-0.53	0.66	1.19	1.35	38.45	0.00
USDT in UA	0.17	0.09	0.17	0.00	1.00	1.00	1.17	10.22	0.00
BTC in UA	0.31	0.12	0.28	0.00	0.98	0.98	1.68	4.30	0.00
ETH in UA	0.31	0.18	0.26	0.00	0.98	0.98	0.93	0.70	0.01
Binance in UA	0.14	0.08	0.12	0.04	1.00	0.96	3.70	26.12	0.00
USD in UA	0.20	0.09	0.18	0.06	1.00	0.94	3.92	24.94	0.00
USDT in US	0.09	0.05	0.08	0.00	1.00	1.00	8.82	128.59	0.00
BTC in US	0.14	0.11	0.10	0.01	1.00	0.99	2.45	9.16	0.00
ETH in US	0.18	0.13	0.14	0.03	1.00	0.97	1.72	4.21	0.01
Binance in US	0.07	0.09	0.05	0.00	1.00	1.00	4.68	33.26	0.00
USD in US	0.40	0.17	0.46	0.09	1.00	0.91	-0.23	-0.85	0.01
USDT in WO	0.12	0.06	0.12	0.00	1.00	1.00	6.27	79.35	0.00
BTC in WO	0.13	0.10	0.11	0.02	1.00	0.98	2.91	15.21	0.00
ETH in WO	0.11	0.09	0.08	0.01	1.00	0.99	3.36	21.40	0.00
Binance in WO	0.16	0.13	0.11	0.00	0.85	0.85	1.88	4.67	0.01
USD in WO	0.27	0.10	0.29	0.09	1.00	0.91	0.85	4.48	0.00
Controls - Daily Changes in:									
RUB\USD	0.02	0.57	-0.00	-0.93	14.29	15.23	24.72	616.06	0.02
S&P500	0.00	0.01	0.00	-0.04	0.06	0.10	-0.10	1.84	0.00
Gold	0.00	0.01	0.00	-0.05	0.03	0.08	-0.30	2.05	0.00
Oil	0.00	0.02	0.00	-0.12	0.08	0.20	-0.28	1.23	0.00
VIX	0.00	0.07	-0.01	-0.20	0.62	0.81	1.61	9.09	0.00

Table 3: Panel B: Descriptive statistics for USDT and ETH metrics on Ethereum blockchain.

This table reports descriptive statistics for daily changes in USDT and Ether (ETH) metrics on the Ethereum network. The USDT metrics include daily changes in: USDT price, number of active, receiver and sender addresses, USDT share held in different address balance brackets (USD), total number of addresses holding any positive amount of USDT, adjusted NVT ratio, USDT supply held by top 100, top 10%, and top 1% addresses, and transaction volumes by size. The ETH metrics include daily changes in: ETH price, number of active, receiver and sender addresses, ETH supply in different address balance brackets, total number of addresses holding any positive amount of ETH, ETH supply held by top percentile addresses, and transaction volumes by size. Prices and transaction volumes are in USD; ETH supply and address brackets in thousands; USDT supply in millions.

	mean	sd	median	min	max	range	skew	kurtosis	se
USDT Metrics on Ethereum - Daily Changes in:									
USDT Price	-0.00	0.00	-0.00	-0.00	0.00	0.01	-1.13	25.71	0.00
USDT Active Addr.	3.9K	14.4K	1.6K	-51.7K	168.1K	219.9K	2.61	28.05	0.6K
USDT Receiver Addr.	3.2K	9.0K	1.1K	-31.6K	51.2K	82.8K	0.93	2.26	0.4K
USDT Sender Addr.	2.4K	11.7K	0.9K	-49.3K	167.6K	216.9K	4.48	63.61	0.5K
Perc. USDT in Addr. $\geq$ 1M USD	-0.03	0.33	0.01	-3.71	0.96	4.67	-3.03	27.40	0.01
Perc. USDT in Addr. [100K,1M) USD	-0.00	0.25	0.01	-2.85	0.76	3.61	-2.97	29.39	0.01
Perc. USDT in Addr. [10K,100K) USD	0.01	0.10	0.01	-0.93	0.39	1.33	-1.86	16.48	0.00
Perc. USDT in Addr. [1K,10K) USD	0.00	0.03	0.00	-0.25	0.12	0.37	-1.97	14.45	0.00
Perc. USDT in Addr. [100,1K) USD	-0.00	0.02	0.00	-0.19	0.07	0.26	-4.53	41.33	0.00
Perc. USDT in Addr. [10,100) USD	0.00	0.01	0.00	-0.02	0.02	0.05	-0.05	3.04	0.00
Perc. USDT in Addr. [1,10) USD	0.00	0.00	0.00	-0.00	0.00	0.00	-0.01	3.66	0.00
Adjusted NVT (Free Float)	-2.31	5.80	-0.40	-32.75	6.69	39.44	-2.33	5.64	0.23
USDT Supply Held by Top 100 Addr.	15.5M	260.9M	-7.3M	-2.07B	2.38B	4.45B	2.95	31.29	10.4M
USDT Supply Held by Top 10% Addr.	35.4M	231.7M	-0.4M	-2.00B	2.40B	4.40B	4.80	53.07	9.2M
USDT Supply Held by Top 1% Addr.	31.9M	232.3M	-3.0M	-2.01B	2.40B	4.40B	4.71	51.92	9.2M
USDT Transactions $\geq$ 100M NUs	53.4M	1.34B	0.00	-17.51B	9.91B	27.42B	-2.47	63.27	53.3M
USDT Transactions $\geq$ 10M NUs	207.2M	1.61B	120.9M	-17.45B	9.72B	27.17B	-1.47	32.04	64.1M
USDT Transactions $\geq$ 1M NUs	444.0M	2.03B	213.6M	-17.51B	10.95B	28.46B	-0.52	16.14	80.7M
USDT Transactions $\geq$ 100K NUs	591.2M	2.33B	222.2M	-17.76B	12.68B	30.44B	-0.21	11.50	92.3M
USDT Transactions $\geq$ 100 NUs	19.1K	191.3K	13.8K	-1.01M	2.61M	3.61M	4.18	59.14	7.6K
USDT Transactions $\geq$ 10K NUs	7.7M	21.8M	2.8M	-67.7M	77.6M	145.3M	0.53	0.57	0.9M
USDT Transactions $\geq$ 1K NUs	0.7M	4.5M	0.4M	-39.4M	53.8M	93.2M	2.76	51.92	0.2M
USDT Transactions $\leq$ 500 NUs	0.3M	2.3M	0.2M	-11.1M	44.2M	55.3M	10.56	199.96	0.09M
ETH Metrics on Ethereum - Daily Changes in:									
ETH Price	0.69	111.4	2.18	-832.5	520.6	1353.1	-0.63	6.78	4.4
ETH Active Addr.	1.8K	72.0K	2.7K	-649.5K	976.7K	1.63M	3.17	74.52	2.9K
ETH Receiver Addr.	-0.8K	27.0K	0.6K	-177.2K	137.6K	314.8K	-0.70	6.83	1.1K
ETH Sender Addr.	2.1K	70.7K	1.4K	-652.7K	980.4K	1.63M	3.32	82.26	2.8K
ETH Supply $\geq$ 10M USD	-20.2K	313.0K	-21.9K	-1.63M	1.44M	3.07M	-0.57	4.53	12.4K
ETH Supply [1M,10M) USD	-4.4K	90.4K	-4.0K	-423.7K	600.2K	1.02M	0.88	7.70	3.6K
ETH Supply [100K,1M) USD	-0.5K	83.3K	-2.2K	-399.5K	435.7K	835.2K	0.56	5.77	3.3K
ETH Supply [10K,100K) USD	1.2K	76.8K	0.4K	-334.9K	534.3K	869.2K	0.64	5.36	3.0K
ETH Supply [1K,10K) USD	0.7K	56.0K	0.6K	-248.3K	319.9K	568.2K	0.34	4.29	2.2K
ETH Supply [100,1K) USD	62.3	23.4K	-0.2K	-219.4K	110.6K	330.0K	-1.68	21.67	0.9K
ETH Supply [10,100) USD	343.1	9.7K	-26.6	-42.7K	48.7K	91.4K	0.31	3.88	0.4K
ETH Supply [1,10) USD	-12.7	2.5K	0.6	-12.7K	14.4K	27.1K	0.27	5.04	0.1K
ETH Transactions $\geq$ 100M NUs	29.9K	1.78M	0.00	-28.0M	24.4M	52.4M	-1.92	154.51	70.9K
ETH Transactions $\geq$ 10M NUs	130.3K	1.91M	110.1K	-28.1M	24.6M	52.7M	-1.56	121.12	75.7K
ETH Transactions $\geq$ 1M NUs	217.8K	2.09M	147.4K	-28.3M	24.9M	53.2M	-1.14	88.11	82.9K
ETH Transactions $\geq$ 100K NUs	257.8K	2.22M	147.6K	-28.6M	24.9M	53.5M	-0.89	71.62	88.3K
ETH Transactions $\geq$ 100 NUs	239.4	2.1K	205.6	-30.6K	28.1K	58.7K	-1.24	121.64	0.1K
ETH Transactions $\geq$ 10K NUs	5.6K	30.4K	3.4K	-205.2K	352.5K	557.7K	2.21	32.04	1.2K
ETH Transactions $\geq$ 1K NUs	1.6K	14.5K	1.1K	-203.1K	212.4K	415.5K	2.33	149.01	0.6K
ETH Transactions $\leq$ 500 NUs	1.0K	8.0K	578.0	-70.9K	142.2K	213.1K	8.23	164.89	0.3K

Table 3: Panel C: Descriptive statistics for USDT metrics on Tron blockchain, and for Bitcoin (BTC) and Binance on-chain metrics.

This table reports descriptive statistics for daily changes in USDT metrics on the Tron network, Bitcoin (BTC) metrics, and Binance on-chain transaction metrics. The USDT metrics include daily changes in: USDT price, number of active, receiver and sender addresses, USDT share held in different address balance brackets (USD), total number of addresses holding any positive amount of USDT, adjusted NVT ratio, USDT supply held by top 100, top 10%, and top 1% addresses (in M), and transaction volumes (in M or B). BTC metrics include similar variables. Binance metrics report transactions to/from Binance.

	mean	sd	median	min	max	range	skew	kurtosis	se
USDT Metrics on Tron - Daily Changes in:									
USDT Price on Tron	-0.00	0.00	-0.00	-0.00	0.00	0.01	-1.13	25.71	0.00
USDT Active Addresses on Tron	20.5K	81.9K	5.9K	-384.2K	485.7K	869.9K	0.87	4.82	3.3K
USDT Receiver Addresses on Tron	18.4K	56.6K	4.3K	-129.8K	239.6K	369.4K	1.46	2.31	2.2K
USDT Sender Addresses on Tron	12.4K	67.8K	4.6K	-384.8K	454.5K	839.3K	0.59	9.73	2.7K
Perc. USDT in Addr. $\geq$ 1M USD	0.00	1.01	0.06	-14.26	1.87	16.13	-9.19	107.10	0.04
Perc. USDT in Addr. [100K,1M) USD	0.03	0.72	0.07	-8.91	2.42	11.33	-7.94	84.55	0.03
Perc. USDT in Addr. [10K,100K) USD	0.02	0.38	0.04	-5.86	0.86	6.72	-8.42	106.27	0.01
Perc. USDT in Addr. [1K,10K) USD	0.00	0.24	0.03	-3.06	0.55	3.61	-8.05	80.34	0.01
Perc. USDT in Addr. [100,1K) USD	0.00	0.04	0.01	-0.51	0.23	0.74	-4.82	53.50	0.00
Perc. USDT in Addr. [10,100) USD	0.00	0.01	0.00	-0.04	0.06	0.10	0.51	9.40	0.00
Perc. USDT in Addr. [1,10) USD	0.00	0.00	0.00	-0.01	0.01	0.02	0.00	7.70	0.00
Adjusted NVT (Free Float)	-0.76	2.18	-0.09	-10.65	6.26	16.91	-0.87	1.14	0.09
USDT Supply Held by Top 100 Addr.	14.8M	723.0M	-61.7M	-3.22B	10.53B	13.75B	6.11	79.35	28.7M
USDT Supply Held by Top 10% Addr.	51.0M	394.4M	-1.7M	-3.10B	2.00B	5.11B	-0.96	20.58	15.7M
USDT Supply Held by Top 1% Addr.	49.8M	412.4M	-9.8M	-3.11B	2.71B	5.82B	-0.46	20.19	16.4M
USDT Transactions $\geq$ 100M NUs	89.6M	2.15B	0.00	-13.32B	26.84B	40.16B	2.80	43.98	85.6M
USDT Transactions $\geq$ 10M NUs	170.2M	2.40B	100.2M	-14.27B	31.42B	45.69B	3.25	49.94	95.4M
USDT Transactions $\geq$ 1M NUs	327.6M	2.73B	148.3M	-14.15B	33.92B	48.07B	2.77	40.39	108.3M
USDT Transactions $\geq$ 100K NUs	644.4M	3.20B	218.6M	-17.17B	34.83B	52.00B	1.90	23.75	127.1M
USDT Transactions $\geq$ 10K NUs	0.41M	1.91M	0.15M	-9.25M	10.88M	20.13M	0.50	3.84	0.08M
USDT Transactions $\geq$ 1K NUs	36.3M	152.6M	5.3M	-1.00B	912.7M	1.92B	0.68	10.22	6.1M
USDT Transactions $\geq$ 500 NUs	4.9M	22.3M	1.3M	-133.9M	157.8M	291.7M	0.94	8.29	0.89M
USDT Transactions $\leq$ 500 NUs	2.5M	13.5M	0.5M	-127.2M	163.1M	290.3M	1.76	47.14	0.54M
Bitcoin BTC Metrics - Daily Changes in:									
BTC Price	6.16	1432.1	-18.61	-5639.8	7118.9	12758.7	0.02	2.90	56.9
BTC Active Addresses	28.4K	99.0K	15.2K	-208.8K	341.6K	550.4K	0.38	-0.08	3.9K
BTC Receiver Addresses	24.8K	60.4K	20.9K	-209.4K	224.7K	434.1K	0.16	0.49	2.4K
BTC Sender Addresses	14.6K	89.2K	5.0K	-235.4K	298.1K	533.5K	0.29	-0.19	3.5K
BTC Share in Addr. $\geq$ 10M USD	101.2	64.7K	-1.4K	-323.8K	245.9K	569.6K	-0.47	5.98	2.6K
BTC Share in Addr. [1M,10M) USD	-2.5K	220.4K	-98.0	-1.68M	1.67M	3.35M	-0.60	53.39	8.8K
BTC Share in Addr. [100K,1M) USD	2.6K	236.5K	0.5K	-1.79M	1.81M	3.60M	0.67	53.77	9.4K
BTC Share in Addr. [10K,100K) USD	0.53K	19.0K	0.58K	-96.3K	101.4K	197.7K	0.39	9.41	0.76K
BTC Share in Addr. [1K,10K) USD	0.20K	11.1K	0.45K	-49.5K	73.5K	123.0K	0.59	7.42	0.44K
BTC Share in Addr. [100,1K) USD	39.2	2.8K	90.0	-10.5K	16.5K	27.0K	0.71	5.99	0.11K
BTC Share in Addr. [10,100) USD	2.32	531.1	5.22	-2217.8	3395.1	5612.9	0.90	7.30	21.1
BTC Share in Addr. [1,10) USD	0.43	50.8	1.10	-238.5	320.0	558.6	0.75	6.69	2.0
BTC Transactions $\geq$ 100M NUs	31.8K	478.6K	7.3K	-7.79M	5.16M	12.95M	-4.29	134.04	19.0K
BTC Transactions $\geq$ 10M NUs	71.7K	629.1K	22.1K	-7.34M	5.37M	12.71M	-0.17	44.69	25.0K
BTC Transactions $\geq$ 1M NUs	89.6K	649.6K	28.2K	-7.34M	5.43M	12.77M	0.03	40.21	25.8K
BTC Transactions $\geq$ 100K NUs	94.7K	653.9K	29.9K	-7.34M	5.45M	12.79M	0.08	39.50	26.0K
BTC Transactions $\geq$ 10K NUs	2.83	16.9	2.59	-114.9	80.3	195.2	-1.17	11.63	0.67
BTC Transactions $\geq$ 1K NUs	436.2	990.0	166.0	-2320.7	4099.9	6420.6	0.93	0.81	39.3
BTC Transactions $\geq$ 500 NUs	65.5	148.2	40.3	-485.9	605.2	1091.1	0.52	0.63	5.9
BTC Transactions $\leq$ 500 NUs	30.4	73.7	19.9	-266.3	294.8	561.1	0.33	0.84	2.9
Binance On-chain Metrics - Daily Changes in:									
BTC Transactions to Binance	2.7K	5.3K	2.2K	-18.8K	33.8K	52.6K	1.02	4.57	0.21K
BTC Transactions from Binance	0.90K	3.6K	108.0	-23.4K	26.9K	50.3K	0.92	11.22	0.14K
ETH Transactions to Binance	0.20K	2.8K	6.00	-9.3K	45.2K	54.6K	12.45	189.72	0.11K
ETH Transactions from Binance	0.31K	3.4K	353.5	-27.1K	25.1K	52.1K	-0.49	18.96	0.14K

Table 4: Parallel Trends Regression Between Russia and the World.

The table presents the results of the parallel trends regression analysis comparing the relationship between search interest in the keywords *USDT*, *BTC*, *ETH*, *Binance*, and *USD* in Russia and the world. The dummy *Russia* is equal to one for observations related to Russian search interest; otherwise, it is zero. This variable captures systematic differences between Russia and the world. The dummy variables *Jan. 2022*, *Dec. 2021*, and *Nov. 2021* are equal to one if the observations relate to the respective periods; otherwise, they are zero. They capture changes over the specified months that affect both groups in a similar way. The interaction terms $Russia \times Jan. 2022$, $Russia \times Nov. 2021$, and $Russia \times Dec. 2021$ indicate whether there is a parallel trend between the groups. ***, ** and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

	<i>Dependent variables:</i>				
	Online Search Interest in:				
	USDT	BTC	ETH	Binance	USD
Russia	-0.08*** (0.02)	-0.08*** (0.03)	-0.01 (0.03)	0.02 (0.03)	-0.13*** (0.03)
Jan. 2022	-0.07 (0.05)	-0.09 (0.09)	-0.06 (0.09)	0.01 (0.10)	-0.10 (0.11)
Nov. 2021	-0.02 (0.02)	-0.12*** (0.03)	-0.18*** (0.03)	-0.05** (0.02)	-0.13*** (0.03)
Dec. 2021	-0.05*** (0.02)	-0.12*** (0.03)	-0.22*** (0.03)	-0.08*** (0.03)	-0.11*** (0.03)
Russia $\times$ Jan. 2022	0.07 (0.07)	0.04 (0.13)	0.07 (0.13)	0.09 (0.14)	0.08 (0.15)
Russia $\times$ Nov. 2021	0.01 (0.02)	0.03 (0.04)	0.03 (0.04)	0.02 (0.04)	0.03 (0.05)
Russia $\times$ Dec. 2021	0.03 (0.02)	0.03 (0.04)	0.08* (0.04)	0.04 (0.04)	0.01 (0.05)
Constant	0.20*** (0.01)	0.25*** (0.02)	0.36*** (0.02)	0.30*** (0.02)	0.31*** (0.02)
Observations	122	122	122	122	122
R ²	0.41	0.33	0.50	0.35	0.37
Adjusted R ²	0.37	0.29	0.47	0.32	0.33

Table 5: Short Term DiD Between Russia and the World.

The table presents the results of the DiD regression analysis comparing the search interest in the selected keywords in Russia and the world, before Russia's invasion of Ukraine on February 24, 2022 and in the 6 months following the event. The dummy *Russia* is equal to one for observations related to Russian search interest; otherwise, it is zero. This variable captures systematic differences between Russia and the world. The dummy variable *Post* is equal to one for all observations in the six months after February 24, 2022. This variable captures changes over time that affect both groups in a similar way. The interaction term *Russia* $\times$ *Post* indicates whether there is any post-invasion difference in search interest between the groups. In some specifications, we include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

<i>Dependent variable:</i>					
Online Search Interest in:					
	USDT	USDT	USDT	USDT	USDT
Russia	0.02** (0.01)	0.02*** (0.01)	-0.02*** (0.01)	-0.02*** (0.01)	-0.02*** (0.01)
Post		0.15*** (0.01)	0.07*** (0.01)	0.05** (0.02)	0.05** (0.02)
Russia $\times$ Post			0.16*** (0.01)	0.16*** (0.01)	0.16*** (0.01)
Constant	0.13*** (0.01)	0.09*** (0.005)	0.11*** (0.005)	0.11 (0.005)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.01	0.37	0.48	0.48	0.28
Adjusted R ²	0.01	0.37	0.48	0.527	0.31
	BTC	BTC	BTC	BTC	BTC
Russia	-0.04*** (0.01)	-0.04*** (0.01)	-0.05*** (0.01)	-0.05*** (0.01)	-0.05*** (0.01)
Post		0.001 (0.01)	-0.01 (0.01)	-0.09*** (0.02)	-0.09*** (0.02)
Russia $\times$ Post			0.03 (0.02)	0.03 (0.02)	0.03 (0.02)
Constant	0.18*** (0.01)	0.18*** (0.01)	0.18*** (0.01)	0.18 (0.01)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.03	0.03	0.04	0.04	0.04
Adjusted R ²	0.03	0.03	0.03	0.03	0.21
	ETH	ETH	ETH	ETH	ETH
Russia	0.10*** (0.01)	0.10*** (0.01)	0.08*** (0.01)	0.08*** (0.01)	0.08*** (0.01)
Post		0.03*** (0.01)	-0.01 (0.01)	-0.14*** (0.02)	-0.14*** (0.02)
Russia $\times$ Post			0.08*** (0.02)	0.07*** (0.02)	0.08*** (0.01)
Constant	0.16*** (0.01)	0.15*** (0.01)	0.16*** (0.01)	0.16*** (0.13)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.22	0.23	0.25	0.26	0.36
Adjusted R ²	0.21	0.22	0.25	0.35	0.40
	Binance	Binance	Binance	Binance	Binance
Russia	-0.10*** (0.01)	-0.10*** (0.01)	-0.13*** (0.01)	-0.13*** (0.01)	-0.13*** (0.01)
Post		0.01 (0.01)	-0.07*** (0.02)	-0.10*** (0.03)	-0.10*** (0.03)
Russia $\times$ Post			0.15*** (0.02)	0.15*** (0.02)	0.15*** (0.02)
Constant	0.25*** (0.01)	0.25*** (0.01)	0.26*** (0.01)	-0.26*** (0.01)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.14	0.14	0.19	0.20	0.21
Adjusted R ²	0.13	0.13	0.19	0.20	0.25
	USD	USD	USD	USD	USD
Russia	-0.05*** (0.01)	-0.05*** (0.01)	-0.07*** (0.01)	-0.07*** (0.01)	-0.07*** (0.01)
Post		0.21*** (0.01)	0.15*** (0.01)	0.02 (0.02)	0.02 (0.02)
Russia $\times$ Post			0.12*** (0.02)	0.12*** (0.02)	0.12*** (0.02)
Constant	0.21*** (0.01)	0.16*** (0.01)	0.18*** (0.01)	0.18*** (0.01)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.03	0.45	0.48	0.49	0.23
Adjusted R ²	0.03	0.45	0.48	0.21	0.26

Table 6: Long-Term DiD Between Russia and the World.

The table presents the results of the DiD regression analysis comparing the search interest in the selected keywords in Russia and the world, before and after the Russia's invasion of Ukraine on February 24, 2022. The dummy *Russia* is equal to one for observations related to Russian search interest; otherwise, it is zero. This variable captures systematic differences between Russia and the world. The dummy variable *Post* is equal to one for all observations after February 24, 2022. The interaction term *Russia* $\times$ *Post* captures whether there is any post-invasion difference in search interest between the two groups. In some specifications, we include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

Dependent variable:					
Online Search Interest in:					
	USDT	USDT	USDT	USDT	USDT
Russia	0.05*** (0.004)	0.05*** (0.004)	-0.02*** (0.01)	-0.02*** (0.01)	-0.02*** (0.01)
Post		0.07*** (0.004)	0.02*** (0.01)	-0.01 (0.01)	0.01 (0.01)
Russia $\times$ Post			0.11*** (0.01)	0.11*** (0.01)	0.11*** (0.01)
Constant	0.12*** (0.003)	0.08*** (0.004)	0.11*** (0.004)	0.11*** (0.004)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,267	1,267	1,268
R ²	0.08	0.26	0.36	0.26	0.25
Adjusted R ²	0.08	0.26	0.26	0.25	0.38
	BTC	BTC	BTC	BTC	BTC
Russia	-0.03*** (0.01)	-0.03*** (0.005)	-0.05*** (0.01)	-0.05*** (0.01)	-0.05*** (0.01)
Post		-0.07*** (0.005)	-0.09*** (0.01)	-0.15*** (0.01)	-0.13*** (0.01)
Russia $\times$ Post			0.03*** (0.01)	0.03*** (0.01)	0.03*** (0.01)
Constant	0.13*** (0.004)	0.18*** (0.004)	0.18*** (0.01)	0.18*** (0.01)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,267	1,267	1,268
R ²	0.02	0.17	0.18	0.18	0.06
Adjusted R ²	0.02	0.17	0.17	0.06	0.23
	ETH	ETH	ETH	ETH	ETH
Russia	0.10*** (0.01)	0.10*** (0.005)	0.08*** (0.01)	0.08*** (0.01)	0.08*** (0.01)
Post		-0.07*** (0.005)	-0.09*** (0.01)	-0.12*** (0.01)	-0.17*** (0.01)
Russia $\times$ Post			0.03*** (0.01)	0.03*** (0.01)	0.03*** (0.01)
Constant	0.11*** (0.004)	0.15*** (0.005)	0.16*** (0.01)	0.16*** (0.01)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,267	1,267	1,268
R ²	0.21	0.32	0.33	0.33	0.39
Adjusted R ²	0.21	0.32	0.33	0.39	0.47
	Binance	Binance	Binance	Binance	Binance
Russia	-0.05*** (0.01)	-0.05*** (0.01)	-0.13*** (0.01)	-0.13*** (0.01)	-0.13*** (0.01)
Post		-0.10*** (0.01)	-0.16*** (0.01)	-0.19*** (0.01)	-0.15*** (0.01)
Russia $\times$ Post			0.13*** (0.01)	0.13*** (0.01)	0.13*** (0.01)
Constant	0.16*** (0.004)	0.22*** (0.01)	0.26*** (0.01)	0.26*** (0.01)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,267	1,267	1,268
R ²	0.04	0.23	0.31	0.31	0.22
Adjusted R ²	0.04	0.22	0.31	0.21	0.30
	USD	USD	USD	USD	USD
Russia	-0.11*** (0.01)	-0.11*** (0.005)	-0.07*** (0.01)	-0.07*** (0.01)	-0.07*** (0.01)
Post		0.13*** (0.01)	0.16*** (0.01)	0.12*** (0.01)	0.10*** (0.01)
Russia $\times$ Post			-0.06*** (0.01)	-0.06*** (0.01)	-0.06*** (0.01)
Constant	0.27*** (0.004)	0.19*** (0.005)	0.18*** (0.01)	0.18*** (0.01)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,267	1,267	1,268
R ²	0.20	0.47	0.48	0.48	0.32
Adjusted R ²	0.19	0.47	0.48	0.32	0.40

Table 7: Dynamic DiD Between Russia and the World.

The table presents the results of the dynamic DiD regression analysis comparing the search interest in the selected keywords in Russia and the world, before Russia’s invasion of Ukraine on February 24, 2022 and in the two years following the event. The dummy *Russia* is equal to one for observations related to Russian search interest; otherwise, it is zero. This variable captures systematic differences between Russia and the world. The dummy variables *Year 2022* and *Year 2023* are equal to one for all observations in the respective years; otherwise, they are equal to zero. These dummy variables capture changes over the specified years that affect both groups (Russia and the world) equally. The interaction terms *Russia* $\times$ *Year 2022* and *Russia* $\times$ *Year 2023* indicate whether there are any post-invasion differences in the search interest between the groups over the two years. We include controls for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

	Dependent variable				
	Online Search Interest in:				
	USDT	BTC	ETH	Binance	USD
Russia	0.01 (0.01)	-0.04*** (0.01)	0.09*** (0.01)	-0.13*** (0.01)	-0.07*** (0.01)
Year 2022	0.05*** (0.01)	-0.04*** (0.01)	0.01 (0.01)	-0.09*** (0.01)	0.16*** (0.01)
Year 2023	-0.02*** (0.01)	-0.10*** (0.01)	-0.09*** (0.01)	-0.19*** (0.01)	0.18*** (0.01)
Russia $\times$ Year 2022	0.08*** (0.01)	0.02 (0.01)	0.02 (0.01)	0.11*** (0.01)	-0.01 (0.01)
Russia $\times$ Year 2023	0.09*** (0.01)	0.03*** (0.01)	0.01 (0.01)	0.13*** (0.01)	-0.11*** (0.01)
Constant	0.10*** (0.005)	0.18*** (0.01)	0.13*** (0.01)	0.26*** (0.01)	0.16*** (0.01)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	1,267	1,267	1,267	1,267	1,267
R ²	0.35	0.19	0.39	0.33	0.54
Adjusted R ²	0.34	0.18	0.38	0.32	0.54

Table 8: Regressions of USDT on-chain metrics (for Ethereum blockchain) on changes in *USDT* search interest in Russia.

This table presents the results of regression analyses examining the relationship between daily changes in USDT on-chain metrics on the Ethereum blockchain (dependent variables) and daily changes in online search interest in the *USDT* keyword by the Russian public. The sample period goes from February 24th, 2022 (invasion of Ukraine), to the end of 2023. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. We include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. Addr. stands for cryptocurrency addresses and NUs for native units. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Var. (Changes in:)	Independent Var. (Changes in:)	Coefficient	Std. Error	t-value	p-value	Significance	Y/M FE	Controls
USDT Price on Ethereum	USDT Interest in Russia	-0.0004	0.0003	-1.09	0.28		✓	✓
Addresses in thousands:								
USDT Active Addresses on Ethereum	USDT Interest in Russia	8.76	12.29	0.71	0.48		✓	✓
USDT Receiver Addresses on Ethereum	USDT Interest in Russia	3.58	7.27	0.49	0.62		✓	✓
USDT Sender Addresses on Ethereum	USDT Interest in Russia	9.08	9.91	0.92	0.36		✓	✓
USDT Share in Addresses (USD):								
Perc. USDT in Addr. $\geq$ 10M USD	USDT Interest in Russia	0.0021	0.0058	0.37	0.71		✓	✓
Perc. USDT in Addr. [1M,10M) USD	USDT Interest in Russia	-0.15	0.16	-0.88	0.38		✓	✓
Perc. USDT in Addr. [100K,1M) USD	USDT Interest in Russia	-0.16	0.13	-1.21	0.23		✓	✓
Perc. USDT in Addr. [10K,100K) USD	USDT Interest in Russia	-0.01	0.05	-0.20	0.85		✓	✓
Perc. USDT in Addr. [1K,10K) USD	USDT Interest in Russia	0.04	0.01	3.50	0.00	***	✓	✓
Perc. USDT in Addr. [100,1K) USD	USDT Interest in Russia	0.00	0.01	0.00	1.00		✓	✓
Perc. USDT in Addr. [10,100) USD	USDT Interest in Russia	0.0010	0.0043	0.23	0.81		✓	✓
Perc. USDT in Addr. [1,10) USD	USDT Interest in Russia	0.0001	0.0004	0.23	0.82		✓	✓
Network Value to Transaction Ratio:								
USDT Adjusted NVT (Free Float)	USDT Interest in Russia	-0.20	0.09	-2.29	0.02	**	✓	✓
Whale Concentration in millions:								
USDT Supply Held by Top 100 Addr.	USDT Interest in Russia	73.09	154.07	0.47	0.64		✓	✓
USDT Supply Held by Top 10% Addr.	USDT Interest in Russia	-95.49	124.08	-0.77	0.44		✓	✓
USDT Supply Held by Top 1% Addr.	USDT Interest in Russia	-90.11	124.82	-0.72	0.47		✓	✓
Large Transactions in millions:								
USDT Transactions Valued $\geq$ 100M NUs	USDT Interest in Russia	1038.63	603.05	1.72	0.09	*	✓	✓
USDT Transactions Valued $\geq$ 10M NUs	USDT Interest in Russia	2380.17	837.97	2.84	0.00	***	✓	✓
USDT Transactions Valued $\geq$ 1M NUs	USDT Interest in Russia	4596.17	1216.03	3.78	0.00	***	✓	✓
USDT Transactions Valued $\geq$ 100K NUs	USDT Interest in Russia	5545.49	1431.94	3.87	0.00	***	✓	✓
Small Transactions in millions:								
USDT Transactions Valued $\leq$ 100 NUs	USDT Interest in Russia	0.15	0.16	0.91	0.36		✓	✓
USDT Transactions Valued $\leq$ 10K NUs	USDT Interest in Russia	23.67	15.15	1.56	0.12		✓	✓
USDT Transactions Valued $\leq$ 1K NUs	USDT Interest in Russia	1.01	3.81	0.27	0.79		✓	✓
USDT Transactions Valued $\leq$ 500 NUs	USDT Interest in Russia	2.24	2.11	1.06	0.29		✓	✓

Table 9: Regressions of USDT on-chain metrics (for Tron blockchain) on changes in *USDT* search interest in Russia. This table presents the results of regression analyses examining the relationship between daily changes in USDT on-chain metrics on the Tron blockchain (dependent variables) and daily changes in online search interest in the *USDT* keyword by the Russian public. The sample period goes from February 24th, 2022 (invasion of Ukraine), to the end of 2023. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. We include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. Addr. stands for cryptocurrency addresses and NUs for native units. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Var. (Changes in:)	Independent Var. (Changes in:)	Coefficient	Std. Error	t-value	p-value	Significance	Y/M FE	Controls
USDT Price on Tron	USDT Interest in Russia	-0.0004	0.0003	-1.09	0.28		✓	✓
Addresses in thousands:								
USDT Active Addresses on Tron	USDT Interest in Russia	69.30	67.08	1.03	0.30		✓	✓
USDT Receiver Addresses on Tron	USDT Interest in Russia	73.78	47.07	1.57	0.12		✓	✓
USDT Sender Addresses on Tron	USDT Interest in Russia	30.24	55.73	0.54	0.59		✓	✓
USDT Share in Addresses (USD):								
Perc. USDT in Addr. $\geq 10M$ USD	USDT Interest in Russia	-0.0014	0.0065	-0.21	0.83		✓	✓
Perc. USDT in Addr. [1M,10M) USD	USDT Interest in Russia	0.22	0.17	1.26	0.21		✓	✓
Perc. USDT in Addr. [100K,1M) USD	USDT Interest in Russia	0.27	0.20	1.35	0.18		✓	✓
Perc. USDT in Addr. [10K,100K) USD	USDT Interest in Russia	0.14	0.10	1.51	0.13		✓	✓
Perc. USDT in Addr. [1K,10K) USD	USDT Interest in Russia	0.03	0.12	0.22	0.83		✓	✓
Perc. USDT in Addr. [100,1K) USD	USDT Interest in Russia	0.07	0.03	2.26	0.02	**	✓	✓
Perc. USDT in Addr. [10,100) USD	USDT Interest in Russia	0.01	0.01	1.30	0.19		✓	✓
Perc. USDT in Addr. [1,10) USD	USDT Interest in Russia	0.00	0.00	0.54	0.59		✓	✓
Network Value to Transaction Ratio:								
USDT Adjusted NVT (Free Float)	USDT Interest in Russia	-0.20	0.05	-4.40	0.00	***	✓	✓
Whale Concentration in millions:								
USDT Supply Held by Top 100 Addr.	USDT Interest in Russia	-636.73	283.59	-2.25	0.03	**	✓	✓
USDT Supply Held by Top 10% Addr.	USDT Interest in Russia	-616.88	249.75	-2.47	0.01	***	✓	✓
USDT Supply Held by Top 1% Addr.	USDT Interest in Russia	-622.22	252.14	-2.47	0.01	***	✓	✓
Large Transactions in millions:								
USDT Transactions Valued $\geq 100M$ NUs	USDT Interest in Russia	6304.95	1495.79	4.22	0.00	***	✓	✓
USDT Transactions Valued $\geq 10M$ NUs	USDT Interest in Russia	7048.92	1687.10	4.18	0.00	***	✓	✓
USDT Transactions Valued $\geq 1M$ NUs	USDT Interest in Russia	7714.34	1862.18	4.14	0.00	***	✓	✓
USDT Transactions Valued $\geq 100K$ NUs	USDT Interest in Russia	8271.43	2239.93	3.69	0.00	***	✓	✓
Small Transactions in millions:								
USDT Transactions Valued ≤ 100 NUs	USDT Interest in Russia	1.87	1.57	1.19	0.23		✓	✓
USDT Transactions Valued $\leq 10K$ NUs	USDT Interest in Russia	191.51	113.39	1.69	0.09	*	✓	✓
USDT Transactions Valued $\leq 1K$ NUs	USDT Interest in Russia	25.34	18.73	1.35	0.18		✓	✓
USDT Transactions Valued ≤ 500 NUs	USDT Interest in Russia	4.16	11.83	0.35	0.73		✓	✓

Table 10: Regressions of Bitcoin on-chain metrics on changes in *BTC* search interest.

This table presents the results of regression analyses examining the relationship between daily changes in BTC on-chain metrics on the Bitcoin blockchain and daily changes in search interest in the *BTC* keyword by the Russian public. The sample period goes from February 24th 2022 (invasion of Ukraine) to the end of 2023. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. We include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. Addr. stands for cryptocurrency addresses and NUs for native units. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Var. (Changes in:)	Independent Var. (Changes in:)	Coefficient	Std. Error	t-value	p-value	Significance	Y/M FE	Controls
BTC Price in USD	BTC Interest in Russia	1384.47	553.64	2.50	0.01	*	✓	✓
Addresses in thousands:								
BTC Active Addresses	BTC Interest in Russia	468.76	129.20	3.63	0.00	***	✓	✓
BTC Receiver Addresses	BTC Interest in Russia	309.51	78.81	3.93	0.00	***	✓	✓
BTC Sender Addresses	BTC Interest in Russia	235.28	120.91	1.95	0.05	**	✓	✓
BTC in Addresses (USD):								
BTC Addr. $\geq 10M$ USD	BTC Interest in Russia	-9403.96	56791.62	-0.17	0.87		✓	✓
BTC Addr. [1M,10M) USD	BTC Interest in Russia	5525.09	399401.69	0.01	0.99		✓	✓
BTC Addr. [100K,1M) USD	BTC Interest in Russia	-11219.66	423471.68	-0.03	0.98		✓	✓
BTC Addr. [10K,100K) USD	BTC Interest in Russia	19782.23	14352.39	1.38	0.17		✓	✓
BTC Addr. [1K,10K) USD	BTC Interest in Russia	-5712.80	9997.18	-0.57	0.57		✓	✓
BTC Addr. [100,1K) USD	BTC Interest in Russia	879.70	2032.73	0.43	0.67		✓	✓
BTC Addr. [10,100) USD	BTC Interest in Russia	179.84	395.27	0.45	0.65		✓	✓
BTC Addr. [1,10) USD	BTC Interest in Russia	27.86	33.83	0.82	0.41		✓	✓
Large Transactions in millions:								
BTC Transactions Valued $\geq 100M$ NUs	BTC Interest in Russia	0.82	0.49	1.66	0.10	*	✓	✓
BTC Transactions Valued $\geq 10M$ NUs	BTC Interest in Russia	1.82	0.86	2.12	0.03	**	✓	✓
BTC Transactions Valued $\geq 1M$ NUs	BTC Interest in Russia	2.49	0.89	2.80	0.01	***	✓	✓
BTC Transactions Valued $\geq 100K$ NUs	BTC Interest in Russia	2.63	0.90	2.93	0.00	***	✓	✓
Small Transactions in hundreds:								
BTC Transactions Valued ≤ 100 NUs	BTC Interest in Russia	0.02	0.28	0.07	0.94		✓	✓
BTC Transactions Valued $\leq 10K$ NUs	BTC Interest in Russia	30.90	14.54	2.13	0.03	**	✓	✓
BTC Transactions Valued $\leq 1K$ NUs	BTC Interest in Russia	1.07	2.13	0.50	0.62		✓	✓
BTC Transactions Valued ≤ 500 NUs	BTC Interest in Russia	0.18	1.06	0.17	0.87		✓	✓

Table 11: Regressions of Ether on-chain metrics on changes in *ETH* search interest.

This table presents the results of regression analyses examining the relationship between daily changes in Ether on-chain metrics on the Ethereum blockchain and daily changes in search interest in the *ETH* keyword by the Russian public. The sample period goes from February 24th 2022 (invasion of Ukraine) to the end of 2023. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. We include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. Addr. stands for cryptocurrency addresses and NUs for native units. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Var. (Changes in:)	Independent Var. (Changes in:)	Coefficient	Std. Error	t-value	p-value	Significance	Y/M FE	Controls
ETH Price in USD	ETH Interest in Russia	-35.59	25.03	-1.42	0.16		✓	✓
Addresses in thousands:								
ETH Active Addresses	ETH Interest in Russia	2.57	67.45	0.04	0.97		✓	✓
ETH Receiver Addresses	ETH Interest in Russia	16.62	16.79	0.99	0.32		✓	✓
ETH Sender Addresses	ETH Interest in Russia	-20.73	67.85	-0.31	0.76		✓	✓
ETH in Addresses (USD):								
ETH Addr. ≥ 10 M USD	ETH Interest in Russia	-161026.18	113147.00	-1.42	0.16		✓	✓
ETH Addr. [1M,10M) USD	ETH Interest in Russia	107953.25	50145.73	2.15	0.03	**	✓	✓
ETH Addr. [100K,1M) USD	ETH Interest in Russia	-38232.66	34144.22	-1.12	0.26		✓	✓
ETH Addr. [10K,100K) USD	ETH Interest in Russia	45171.21	21806.64	2.07	0.04	**	✓	✓
ETH Addr. [1K,10K) USD	ETH Interest in Russia	19192.48	25602.02	0.75	0.45		✓	✓
ETH Addr. [100,1K) USD	ETH Interest in Russia	16670.24	14429.80	1.16	0.25		✓	✓
ETH Addr. [10,100) USD	ETH Interest in Russia	1955.07	4470.17	0.44	0.66		✓	✓
ETH Addr. [1,10) USD	ETH Interest in Russia	1564.42	1077.68	1.45	0.15		✓	✓
Network Value to Transaction Ratio:								
ETH Adjusted NVT (Free Float)	ETH Interest in Russia	-0.99	0.34	-2.96	0.00	***	✓	✓
Large Transactions in millions:								
ETH Transactions Valued ≥ 100 M NUs	ETH Interest in Russia	-0.99	0.34	-2.96	0.00	***	✓	✓
ETH Transactions Valued ≥ 10 M NUs	ETH Interest in Russia	-0.32	0.53	-0.59	0.55		✓	✓
ETH Transactions Valued ≥ 1 M NUs	ETH Interest in Russia	1.11	0.72	1.55	0.12		✓	✓
ETH Transactions Valued ≥ 100 K NUs	ETH Interest in Russia	1.69	0.82	2.05	0.04	**	✓	✓
Small Transactions in hundreds:								
ETH Transactions Valued ≤ 100 NUs	ETH Interest in Russia	28.75	18.62	1.54	0.12		✓	✓
ETH Transactions Valued ≤ 10 K NUs	ETH Interest in Russia	412.73	248.67	1.66	0.10	*	✓	✓
ETH Transactions Valued ≤ 1 K NUs	ETH Interest in Russia	86.08	137.18	0.63	0.53		✓	✓
ETH Transactions Valued ≤ 500 NUs	ETH Interest in Russia	47.59	72.82	0.65	0.51		✓	✓

Table 12: Regressions of Binance metrics on changes in *Binance* search interest.

This table presents the results of regression analyses examining the relationship between daily changes in transaction metrics for Bitcoin and Ethereum on the Binance platform, and daily changes in search interest in the *Binance* keyword by the Russian public. The sample period goes from February 24th 2022 (invasion of Ukraine) to the end of 2023. We include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. Addr. stands for cryptocurrency addresses and NUs for native units. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Var. (Changes in:)	Independent Var. (Changes in:)	Coefficient	Std. Error	t-value	p-value	Significance	Y/M FE	Controls
BTC Binance Flows in thousands:								
BTC Transactions to Binance	Binance Interest in Russia	5.53	4.65	1.19	0.23		✓	✓
BTC Transactions from Binance	Binance Interest in Russia	11.68	5.18	2.26	0.02	**	✓	✓
ETH Binance Flows in thousands:								
ETH Transactions to Binance	Binance Interest in Russia	3217.05	4410.00	0.73	0.47		✓	✓
ETH Transactions from Binance	Binance Interest in Russia	8421.02	3246.34	2.59	0.01	***	✓	✓

Table 13: Regressions of USDT on-chain metrics (for Ethereum blockchain) on changes in *USD* search interest. This table presents the results of regression analyses examining the relationship between daily changes in USDT on-chain metrics on the Ethereum blockchain, and daily changes in search interest in the *USD* keyword by the Russian public. The sample period goes from February 24th 2022 (invasion of Ukraine) to the end of 2023. We include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. Addr. stands for cryptocurrency addresses and NUs for native units. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Var. (Changes in:)	Independent Var. (Changes in:)	Coefficient	Std. Error	t-value	p-value	Significance	Y/M FE	Controls
USDT Price on Ethereum	USD Interest in Russia	-0.0004	0.0003	-1.26	0.21		✓	✓
Addresses in thousands:								
USDT Addresses on Ethereum	USD Interest in Russia	15.17	11.26	1.35	0.18		✓	✓
USDT Receiver Addresses on Ethereum	USD Interest in Russia	13.37	6.64	2.01	0.04	**	✓	✓
USDT Sender Addresses on Ethereum	USD Interest in Russia	8.87	9.09	0.98	0.33		✓	✓
USDT Share in Addresses (USD):								
Perc. USDT in Addr. $\geq 10M$ USD	USD Interest in Russia	0.0021	0.0058	0.37	0.71		✓	✓
Perc. USDT in Addr. $[1M,10M)$ USD	USD Interest in Russia	0.02	0.15	0.11	0.91		✓	✓
Perc. USDT in Addr. $[100K,1M)$ USD	USD Interest in Russia	-0.10	0.12	-0.78	0.43		✓	✓
Perc. USDT in Addr. $[10K,100K)$ USD	USD Interest in Russia	0.01	0.04	0.24	0.81		✓	✓
Perc. USDT in Addr. $[1K,10K)$ USD	USD Interest in Russia	0.02	0.01	1.44	0.15		✓	✓
Perc. USDT in Addr. $[100,1K)$ USD	USD Interest in Russia	0.00	0.01	0.16	0.87		✓	✓
Perc. USDT in Addr. $[10,100)$ USD	USD Interest in Russia	-0.0021	0.0039	-0.54	0.59		✓	✓
Perc. USDT in Addr. $[1,10)$ USD	USD Interest in Russia	-0.0002	0.0004	-0.46	0.65		✓	✓
Network Value to Transaction Ratio:								
USDT Adjusted NVT (Free Float)	USD Interest in Russia	-0.06	0.08	-0.76	0.45		✓	✓
Whale Concentration in millions:								
USDT Supply Held by Top 100 Addr.	USD Interest in Russia	18.75	141.44	0.13	0.89		✓	✓
USDT Supply Held by Top 10% Addr.	USD Interest in Russia	-11.64	113.96	-0.10	0.92		✓	✓
USDT Supply Held by Top 1% Addr.	USD Interest in Russia	-23.24	114.63	-0.20	0.84		✓	✓
Large Transactions in millions:								
USDT Transactions Valued $\geq 100M$ NUs	USD Interest in Russia	488.74	555.03	0.88	0.38		✓	✓
USDT Transactions Valued $\geq 10M$ NUs	USD Interest in Russia	1981.71	770.48	2.57	0.01	***	✓	✓
USDT Transactions Valued $\geq 1M$ NUs	USD Interest in Russia	3521.91	1122.34	3.14	0.00	***	✓	✓
USDT Transactions Valued $\geq 100K$ NUs	USD Interest in Russia	4393.59	1320.74	3.33	0.00	***	✓	✓
Small Transactions in millions:								
USDT Transactions Valued ≤ 100 NUs	USD Interest in Russia	0.17	0.15	1.13	0.26		✓	✓
USDT Transactions Valued $\leq 10K$ NUs	USD Interest in Russia	36.80	13.82	2.66	0.01	***	✓	✓
USDT Transactions Valued $\leq 1K$ NUs	USD Interest in Russia	1.46	3.50	0.42	0.68		✓	✓
USDT Transactions Valued ≤ 500 NUs	USD Interest in Russia	1.27	1.94	0.65	0.51		✓	✓

Table 14: Regressions of USDT on-chain metrics (for Tron blockchain) on changes in *USD* search interest. This table presents the results of regression analyses examining the relationship between daily changes in USDT on-chain metrics on the Tron blockchain and daily changes in search interest in the *USD* keyword by the Russian public. The sample period goes from February 24th 2022 (invasion of Ukraine) to the end of 2023. We include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. Addr. stands for cryptocurrency addresses and NUs for native units. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Var. (Changes in:)	Independent Var. (Changes in:)	Coefficient	Std. Error	t-value	p-value	Significance	Y/M FE	Controls
USDT Price on Tron	USD Interest in Russia	-0.0004	0.0003	-1.26	0.21		✓	✓
Addresses in thousands:								
USDT Active Addresses on Tron	USD Interest in Russia	69.01	61.55	1.12	0.26		✓	✓
USDT Receiver Addresses on Tron	USD Interest in Russia	83.47	43.12	1.94	0.05	*	✓	✓
USDT Sender Addresses on Tron	USD Interest in Russia	23.72	51.15	0.46	0.64		✓	✓
USDT Share in Addresses (USD):								
Perc. USDT in Addr. $\geq 10M$ USD	USD Interest in Russia	-0.0014	0.0065	-0.21	0.83		✓	✓
Perc. USDT in Addr. $[1M,10M)$ USD	USD Interest in Russia	0.06	0.16	0.35	0.73		✓	✓
Perc. USDT in Addr. $[100K,1M)$ USD	USD Interest in Russia	-0.10	0.18	-0.56	0.58		✓	✓
Perc. USDT in Addr. $[10K,100K)$ USD	USD Interest in Russia	0.02	0.09	0.27	0.79		✓	✓
Perc. USDT in Addr. $[1K,10K)$ USD	USD Interest in Russia	-0.00	0.11	-0.03	0.97		✓	✓
Perc. USDT in Addr. $[100,1K)$ USD	USD Interest in Russia	0.03	0.03	1.07	0.29		✓	✓
Perc. USDT in Addr. $[10,100)$ USD	USD Interest in Russia	-0.00	0.01	-0.03	0.97		✓	✓
Perc. USDT in Addr. $[1,10)$ USD	USD Interest in Russia	-0.0010	0.0021	-0.45	0.66		✓	✓
Network Value to Transaction Ratio:								
USDT Adjusted NVT (Free Float)	USD Interest in Russia	-0.02	0.04	-0.36	0.72		✓	✓
Whale Concentration in millions:								
USDT Supply Held by Top 100 Addr.	USD Interest in Russia	63.47	261.96	0.24	0.81		✓	✓
USDT Supply Held by Top 10% Addr.	USD Interest in Russia	32.86	231.04	0.14	0.89		✓	✓
USDT Supply Held by Top 1% Addr.	USD Interest in Russia	24.81	233.24	0.11	0.92		✓	✓
Large Transactions in millions:								
USDT Transactions Valued $\geq 100M$ NUs	USD Interest in Russia	1890.71	1401.17	1.35	0.18		✓	✓
USDT Transactions Valued $\geq 10M$ NUs	USD Interest in Russia	1980.96	1580.29	1.25	0.21		✓	✓
USDT Transactions Valued $\geq 1M$ NUs	USD Interest in Russia	2816.48	1741.24	1.62	0.11		✓	✓
USDT Transactions Valued $\geq 100K$ NUs	USD Interest in Russia	3867.06	2082.85	1.86	0.06	*	✓	✓
Small Transactions in millions:								
USDT Transactions Valued ≤ 100 NUs	USD Interest in Russia	1.22	1.44	0.85	0.40		✓	✓
USDT Transactions Valued $\leq 10K$ NUs	USD Interest in Russia	175.40	104.06	1.69	0.09		✓	✓
USDT Transactions Valued $\leq 1K$ NUs	USD Interest in Russia	19.25	17.20	1.12	0.26		✓	✓
USDT Transactions Valued ≤ 500 NUs	USD Interest in Russia	9.61	10.85	0.89	0.38		✓	✓

Table 15: Robustness Check: Placebo DiD between Russia and the World.

The table presents the results of the placebo DiD regression analysis comparing the search interest for the selected keywords in Russia and the world, for the six-month period preceding Russia's invasion of Ukraine (February 24, 2022). The dummy *Russia* is equal to one for observations related to Russian search interest; otherwise, it is zero. This variable captures systematic differences between Russia and the world. The dummy variable *Post* is equal to one for all observations in the six months before February 24, 2022. The interaction term *Russia* $\times$ *Post* captures whether there is any difference in search interest between the two groups (Russia and the world) in the six months before the invasion. We include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

Dependent variable: Online Search Interest in:					
	USDT	BTC	ETH	Binance	USD
Russia	0.01 (0.01)	-0.04*** (0.01)	0.11*** (0.01)	-0.13*** (0.02)	-0.05*** (0.01)
Post	0.06*** (0.01)	-0.08*** (0.01)	0.03** (0.01)	0.04** (0.02)	0.03*** (0.01)
Russia $\times$ Post	-0.05*** (0.01)	-0.01 (0.02)	-0.07*** (0.02)	-0.01 (0.02)	-0.06*** (0.01)
Y/M Fixed Effects	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Observations	461	461	461	461	462
R ²	0.16	0.17	0.23	0.23	0.23
Adjusted R ²	0.14	0.15	0.21	0.22	0.21

Table 16: Robustness Check: RDiT Regressions for Russia.

The table presents the results of a sharp Regression Discontinuity in Time (RDiT) analysis for search interest in the selected keywords. The dummy variable *Post* equals one in the period after Russia's invasion of Ukraine (February 22, 2024); otherwise it is zero. *Days* is a continuous variable representing the days before and after the invasion, and *Russia* $\times$ *Days* is the interaction term. We include controls for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

Dependent variable: Online Search Interest in:					
	USDT	BTC	ETH	Binance	USD
Post	0.29*** (0.02)	0.06*** (0.01)	0.04** (0.02)	0.10*** (0.01)	0.31*** (0.02)
Days	-0.0000 (0.0001)	0.0002** (0.0001)	0.001*** (0.0001)	0.0004*** (0.0001)	0.001*** (0.0001)
Russia $\times$ Days	-0.001*** (0.0002)	-0.001*** (0.0001)	-0.002*** (0.0001)	-0.002*** (0.0001)	-0.002*** (0.0002)
Constant	0.10*** (0.01)	0.12*** (0.01)	0.32*** (0.01)	0.20*** (0.01)	0.16*** (0.02)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	231	231	231	231	231
R ²	0.68	0.22	0.43	0.54	0.70
Adjusted R ²	0.67	0.19	0.41	0.52	0.69

Table 17: Robustness Check: Donut RDiT Regressions for Russia.

The table presents the results of a donut Regression Discontinuity in Time (RDiT) analysis for search interest in selected keywords in Russia, excluding the month before and after Russia's invasion of Ukraine (February 24, 2022) to address potential heaping or anticipation effects. The dummy variable *Post* equals one in the period after the invasion; otherwise, it is zero. *Days* is a continuous variable that represents the number of days before and after the invasion, and *Post* $\times$ *Days* is the interaction term. We include controls for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

Dependent variable:					
	(USDT)	(BTC)	(ETH)	(Binance)	(USD)
Post	0.28*** (0.03)	0.05** (0.02)	0.06** (0.02)	0.03* (0.02)	0.22*** (0.02)
Days	-0.0001 (0.0001)	0.0001 (0.0001)	0.001*** (0.0001)	0.001*** (0.0001)	0.001*** (0.0001)
Post $\times$ Days	-0.001*** (0.0002)	-0.001*** (0.0002)	-0.001*** (0.0002)	-0.001*** (0.0001)	-0.001*** (0.0002)
Constant	0.10*** (0.02)	0.11*** (0.01)	0.29*** (0.02)	0.22*** (0.01)	0.15*** (0.01)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	189	189	189	189	189
R ²	0.61	0.15	0.33	0.44	0.78
Adjusted R ²	0.60	0.12	0.30	0.42	0.78

Table 18: Event Studies: Impact of Sanctions on Cumulative Abnormal Search Interest for the Selected Keywords
This table reports the results of event studies assessing the impact of invasion-related sanctions (independent variables) on the cumulative abnormal interest (CAI) in the keywords *USDT*, *BTC*, *ETH*, *Binance*, and *USD* in Russia. Each row corresponds to a sanction date, and each column shows the estimated coefficient, t statistics, and significance level (***, **, *) for the CAI of the respective keyword. The estimation window spans from 90 days to 10 days before each sanction date; the event window is from 10 days before to 10 days after.

Sanction Date	USDT CAI		BTC CAI		ETH CAI		Binance CAI		USD CAI	
	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat	Coeff	t-stat
2022-02-21	27.99**	2.09	10.37**	1.96	1.8369	0.8551	22.71**	3.13	112.55***	3.28
2022-02-23	114.44**	2.52	17.43**	3.14	3.8922*	1.8076	59.55**	2.84	157.53***	4.15
2022-02-25	149.28***	3.79	27.97***	5.37	7.6271**	2.6329	93.77***	4.41	177.98***	5.25
2022-02-28	187.50***	6.76	24.01***	4.92	8.7392**	3.1200	109.49***	6.23	155.08***	6.07
2022-03-04	161.07***	9.98	17.47***	4.72	12.6105***	5.5977	80.36***	4.78	142.45***	5.36
2022-03-09	140.81***	12.00	12.55***	4.07	8.7082***	3.8068	41.61**	2.33	86.83***	4.06
2022-03-10	128.92***	13.92	9.60**	2.83	8.6426***	3.8149	23.22**	2.80	65.01***	3.97
2022-03-11	118.74***	14.91	6.80***	3.54	3.2156	1.2442	14.15***	3.21	50.09***	5.75
2022-03-14	92.08***	13.77	3.36	1.23	-6.6506	-2.1591	8.52***	3.65	34.92***	4.82
2022-03-15	93.78***	12.54	0.04	0.02	3.3761	0.8332	4.96*	1.81	29.10**	3.04
2022-04-08	49.50***	3.64	-11.81	-7.61	-0.2523	-0.0648	-10.10	-2.51	5.32	0.46
2022-04-21	-8.18	-0.96	-8.52	-5.01	-1.6138	-1.1781	-5.25	-0.41	-35.99	-5.79
2022-06-03	-66.32	-9.81	-13.51	-5.42	-1.3302	-0.3738	-38.41	-18.89	-34.45	-7.89
2022-06-07	-50.70	-6.10	-13.29	-5.05	-0.0388	-0.0099	-31.17	-20.25	-18.01	-3.05
2022-07-01	-33.17	-4.37	-2.24	-0.96	2.2354	0.6796	-24.19	-12.99	21.83*	1.75
2022-07-21	-30.62	-6.49	-6.45	-1.85	0.2896	0.1126	-12.08	-3.84	-14.63	-2.06
2022-07-27	-11.58	-1.61	-9.56	-4.38	-2.9783	-1.3775	-7.58	-2.99	-13.59	-2.43
2022-08-01	-33.14	-8.53	-13.63	-26.70	1.5393	0.7486	-11.96	-4.57	-23.53	-3.72
2022-09-01	-15.19	-2.37	-15.90	-13.44	-3.6640	-1.1501	-3.70	-1.99	-30.17	-6.13
2022-09-12	0.57	0.05	-7.92	-4.90	-0.1197	-0.0438	2.04	0.51	-30.14	-6.13
2022-09-15	-18.51	-2.20	-6.94	-3.75	-4.4020	-1.5470	1.82	0.41	-29.17	-7.70
2022-10-01	3.25	0.48	-8.80	-5.33	-3.6423	-1.3907	8.06	0.77	-20.10	-3.39
2022-10-06	-8.13	-0.92	-9.34	-6.83	-0.5469	-0.3012	9.40	0.84	-19.51	-5.79
2022-10-14	-21.09	-4.16	-9.90	-5.89	11.6700***	4.9918	-9.06	-2.55	-19.78	-4.83
2022-11-14	-14.90	-4.07	1.26	0.92	14.4568***	4.2612	0.17	0.06	-11.93	-3.46
2022-12-06	-21.86	-4.46	-6.67	-4.71	6.1198***	3.7856	-9.48	-13.09	-5.45	-1.65
2022-12-09	-20.54	-5.18	-5.42	-2.59	5.6345*	1.6336	-7.21	-1.89	-6.25	-1.92
2022-12-12	-13.29	-2.16	-4.14	-2.13	5.0845***	4.0284	-4.91	-1.25	-2.26	-0.56
2022-12-16	1.59	0.26	-4.26	-8.76	0.6109	0.3848	-4.09	-2.56	13.94	1.38
2022-12-17	0.33	0.05	-4.57	-10.44	0.8058	0.4846	-3.85	-2.37	18.35*	1.68
2023-02-25	5.81*	1.48	1.52	0.78	-4.1502	-5.2194	-0.14	-0.07	-2.68	-0.77
2023-02-28	9.40**	2.57	-0.29	-0.17	-4.9737	-5.7284	-4.05	-4.49	-7.26	-1.47
2023-03-07	38.92**	2.34	4.76**	2.09	-11.3144	-5.2243	5.02	0.81	-5.72	-1.85
2023-03-14	32.58**	2.72	17.88***	6.68	-13.5787	-6.0276	0.29	0.13	7.97*	1.58
2023-03-31	61.26**	2.65	-3.63	-2.42	-17.1885	-10.9004	-3.89	-3.93	4.07	0.72
2023-04-13	7.00**	2.72	0.36	0.16	-16.2068	-6.5186	0.02	0.01	6.59	1.37
2023-04-14	6.18**	2.71	0.01	0.01	-22.6073	-9.1826	-1.99	-0.78	3.84	1.02
2023-04-24	1.26	0.27	-1.47	-0.77	-21.2918	-9.6621	-2.74	-1.08	-6.77	-2.25
2023-05-20	-39.80	-8.56	-9.40	-4.86	-17.2107	-7.3353	-11.32	-3.59	-9.88	-2.09
2023-05-30	-22.72	-5.12	-3.05	-2.98	52.5983**	2.2008	-5.61	-5.84	1.94	0.43
2023-06-05	-30.25	-7.36	-2.71	-1.94	54.7912**	2.3691	1.84	1.00	-0.34	-0.09
2023-06-23	2.18	0.24	3.96*	1.66	-4.1812	-1.4853	2.64	0.93	12.80**	2.80
2023-07-20	-14.28	-2.20	-1.99	-1.33	-10.1782	-3.0829	3.40**	2.20	-0.24	-0.04
2023-07-28	-10.15	-3.22	-3.88	-1.65	-4.3582	-1.3552	1.96	0.96	3.32	0.64
2023-08-03	3.51	1.00	-4.00	-2.31	-9.3062	-2.5828	1.97	1.20	7.68	1.31
2023-08-22	-3.32	-0.47	-0.79	-0.37	-16.8017	-5.3613	21.19**	2.48	-2.57	-0.51
2023-08-29	-1.39	-0.13	-1.33	-0.61	-15.7160	-12.2958	8.46*	1.74	-2.88	-0.43
2023-09-08	-6.31	-0.92	-3.88	-2.37	-19.8348	-13.6059	-11.45	-7.35	-8.29	-1.49
2023-09-14	-7.34	-1.24	-6.63	-3.24	-17.4255	-4.5878	-10.24	-4.93	-14.46	-2.18
2023-09-24	10.20	1.26	-1.39	-0.86	-10.3325	-3.6447	3.08	0.29	-21.47	-5.58
2023-09-28	8.84	1.26	3.03**	2.46	-4.2785	-2.5279	5.21	0.50	-12.85	-2.37
2023-10-07	-19.20	-6.32	-4.43	-3.65	-1.4942	-0.7148	-16.76	-8.74	-10.47	-2.31
2023-11-03	20.73**	2.99	0.64	0.42	6.7409**	2.7162	-9.83	-4.77	-1.39	-0.34
2023-11-08	4.78	0.52	5.23**	2.63	9.4278***	3.2438	-5.47	-2.85	-5.06	-1.92
2023-11-09	4.95	0.53	6.26***	3.47	13.4234***	4.9973	-5.83	-2.74	-5.47	-2.14
2023-11-17	24.93**	2.97	5.37***	3.56	11.9582*	1.5463	-7.29	-2.47	4.64	0.76
2023-12-06	13.92***	3.79	8.63***	4.65	13.2619**	2.0327	1.25	0.78	6.41	1.08
2023-12-08	16.24***	4.65	4.41**	2.49	15.7965**	2.5500	-0.12	-0.07	0.66	0.21
2023-12-12	16.94**	2.50	4.36**	2.42	12.7069**	2.0549	-3.41	-1.52	-0.28	-0.08
2023-12-18	7.91	0.88	3.35**	1.99	7.0541**	2.0777	-3.83	-3.99	1.70	0.45

Appendix

Table A1: Parallel Trends Regression between Russia and the United States.

The table presents the results of the parallel trends regression analysis comparing the relationship between search interest in the keywords *USDT*, *BTC*, *ETH*, *Binance*, and *USD* in Russia and the United States. The dummy *Russia* is equal to one for observations related to Russian search interest; otherwise, it is zero. This variable captures systematic differences between Russia and the United States. The dummy variables *Jan. 2022*, *Dec. 2021*, and *Nov. 2021* are equal to one if the observations relate to the respective periods; otherwise, they are zero. They capture changes over the specified months that affect both groups in a similar way. The interaction terms *Russia* $\times$ *Jan. 2022*, *Russia* $\times$ *Nov. 2021*, and *Russia* $\times$ *Dec. 2021* indicate whether there is a parallel trend between the groups. ***, ** and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

	Dependent variables: Online Search Interest in:				
	USDT	BTC	ETH	Binance	USD
Russia	0.01 (0.01)	-0.02 (0.02)	-0.16*** (0.02)	0.12*** (0.01)	-0.15*** (0.01)
Jan. 2022	-0.02 (0.03)	-0.06 (0.05)	0.03 (0.06)	-0.01 (0.04)	0.01 (0.04)
Nov. 2021	0.04*** (0.01)	-0.06*** (0.02)	-0.15*** (0.02)	0.05*** (0.01)	-0.09*** (0.01)
Dec. 2021	-0.01 (0.01)	-0.07*** (0.02)	-0.21*** (0.02)	-0.01 (0.01)	-0.12*** (0.01)
Russia $\times$ Jan. 2022	0.01 (0.04)	0.01 (0.08)	0.01 (0.09)	-0.04 (0.05)	-0.03 (0.05)
Russia $\times$ Nov. 2021	-0.06*** (0.01)	-0.04 (0.02)	-0.002 (0.03)	-0.07*** (0.02)	-0.01 (0.02)
Russia $\times$ Dec. 2021	-0.02 (0.01)	-0.02 (0.02)	0.07** (0.03)	-0.04*** (0.02)	0.02 (0.02)
Constant	0.11*** (0.01)	0.20*** (0.01)	0.51*** (0.01)	0.09*** (0.01)	0.33*** (0.01)
Observations	122	122	122	122	122
R ²	0.30	0.43	0.77	0.65	0.86
Adjusted R ²	0.25	0.75	0.67	0.63	0.85

Table A2: Parallel Trends Regression between Russia and Ukraine.

The table presents the results of the parallel trends regression analysis comparing the relationship between search interest in the keywords *USDT*, *BTC*, *ETH*, *Binance*, and *USD* in Russia and Ukraine. The dummy *Russia* is equal to one for observations related to Russian search interest; otherwise, it is zero. This variable captures systematic differences between Russia and Ukraine. The dummy variables *Jan. 2022*, *Dec. 2021*, and *Nov. 2021* are equal to one if the observations relate to the respective periods; otherwise, they are zero. They capture changes over the specified months that affect both groups in a similar way. The interaction terms *Russia* $\times$ *Jan. 2022*, *Russia* $\times$ *Nov. 2021*, and *Russia* $\times$ *Dec. 2021* indicate whether there is a parallel trend between the groups. ***, ** and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

	Dependent variable: Online Search Interest in:				
	USDT	BTC	ETH	Binance	USD
Russia	-0.01 (0.01)	-0.25*** (0.02)	-0.25*** (0.03)	0.05*** (0.01)	-0.12*** (0.01)
Jan. 2022	-0.03 (0.04)	0.01 (0.07)	-0.11 (0.10)	-0.01 (0.04)	-0.05 (0.04)
Nov. 2021	-0.02* (0.01)	-0.14*** (0.02)	-0.25*** (0.03)	-0.01 (0.01)	-0.13*** (0.01)
Dec. 2021	-0.04*** (0.01)	-0.12*** (0.02)	-0.21*** (0.03)	-0.04*** (0.01)	-0.12*** (0.01)
Russia $\times$ Jan. 2022	0.02 (0.06)	-0.06 (0.10)	0.12 (0.14)	-0.04 (0.05)	0.03 (0.05)
Russia $\times$ Nov. 2021	0.01 (0.02)	0.04 (0.03)	0.10** (0.04)	-0.01 (0.02)	0.04* (0.02)
Russia $\times$ Dec. 2021	0.01 (0.02)	0.02 (0.03)	0.06 (0.04)	-0.01 (0.02)	0.02 (0.02)
Constant	0.12*** (0.01)	0.42*** (0.02)	0.59*** (0.02)	0.16*** (0.01)	0.30*** (0.01)
Observations	122	122	122	122	122
R ²	0.09	0.78	0.67	0.37	0.82
Adjusted R ²	0.04	0.77	0.65	0.33	0.81

Table A3: Short Term DiD between Russia and the United States.

The table presents the results of the DiD regression analysis comparing the search interest in the selected keywords in Russia and the United States, before Russia's invasion of Ukraine on February 24, 2022, and in the 6 months following that event. The dummy *Russia* is equal to one for observations related to Russian search interest; otherwise, it is zero. This variable captures systematic differences between Russia and the United States. The dummy variable *Post* is equal to one for all observations in the six months after February 24, 2022. This variable captures changes over time that affect both groups in a similar way. The interaction term *Russia* $\times$ *Post* indicates whether there is any post-invasion difference in the search interest between the groups. In some specifications, we include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

<i>Dependent variables:</i>					
Online Search Interest In:					
	USDT	USDT	USDT	USDT	USDT
Russia	0.05*** (0.01)	0.05*** (0.01)	0.002 (0.01)	0.003 (0.01)	0.003 (0.01)
Post		0.13*** (0.01)	0.03*** (0.01)	0.03*** (0.01)	0.02 (0.01)
Russia $\times$ Post			0.20*** (0.01)	0.20*** (0.01)	0.20*** (0.01)
Constant	0.10*** (0.01)	0.07*** (0.01)	0.09*** (0.005)	0.09*** (0.005)	0.09*** (0.005)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.05	0.34	0.50	0.51	0.39
Adjusted R ²	0.05	0.33	0.50	0.50	0.38
	BTC	BTC	BTC	BTC	BTC
Russia	-0.05*** (0.01)	-0.05*** (0.01)	-0.06*** (0.01)	-0.06*** (0.01)	-0.06*** (0.01)
Post		-0.01 (0.01)	-0.03** (0.02)	-0.03** (0.02)	-0.02 (0.02)
Russia $\times$ Post			0.05** (0.02)	0.05** (0.02)	0.05** (0.02)
Constant	0.19*** (0.01)	0.19*** (0.01)	0.20*** (0.01)	0.20*** (0.01)	0.20*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.04	0.04	0.05	0.05	0.05
Adjusted R ²	0.04	0.04	0.05	0.05	0.05
	ETH	ETH	ETH	ETH	ETH
Russia	-0.01 (0.01)	-0.01 (0.01)	-0.04*** (0.01)	-0.04*** (0.01)	-0.04*** (0.01)
Post		0.01 (0.01)	-0.05*** (0.01)	-0.05*** (0.01)	-0.18*** (0.02)
Russia $\times$ Post			0.11*** (0.02)	0.11*** (0.02)	0.11*** (0.02)
Constant	0.27*** (0.01)	0.26*** (0.01)	0.28*** (0.01)	0.28*** (0.01)	0.28*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.003	0.004	0.05	0.05	0.16
Adjusted R ²	0.001	0.001	0.04	0.04	0.15
	Binance	Binance	Binance	Binance	Binance
Russia	-0.05*** (0.01)	-0.05*** (0.01)	-0.06*** (0.01)	-0.06*** (0.01)	-0.06*** (0.01)
Post		-0.01 (0.01)	-0.03** (0.02)	-0.03** (0.02)	-0.02 (0.02)
Russia $\times$ Post			0.05** (0.02)	0.05** (0.02)	0.05** (0.02)
Constant	0.19*** (0.01)	0.19*** (0.01)	0.20*** (0.01)	0.20*** (0.01)	0.20*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.04	0.04	0.05	0.05	0.05
Adjusted R ²	0.04	0.04	0.05	0.05	0.05
	USD	USD	USD	USD	USD
Russia	0.05*** (0.01)	0.05*** (0.01)	0.01 (0.01)	0.01 (0.01)	0.01 (0.01)
Post		0.01 (0.01)	-0.07*** (0.01)	-0.07*** (0.01)	-0.07*** (0.02)
Russia $\times$ Post			0.15*** (0.02)	0.15*** (0.02)	0.15*** (0.02)
Constant	0.11*** (0.01)	0.11*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.05	0.05	0.14	0.15	0.15
Adjusted R ²	0.05	0.05	0.13	0.14	0.13

Table A4: Short Term DiD between Russia and Ukraine.

The table presents the results of the DiD regression analysis comparing the search interest in the selected keywords in Russia and Ukraine, before Russia's invasion of Ukraine on February 24, 2022, and in the 6 months following that event. The dummy *Russia* is equal to one for observations related to Russian search interest; otherwise, it is zero. This variable captures systematic differences between Russia and Ukraine. The dummy variable *Post* is equal to one for all observations in the six months after February 24, 2022. This variable captures changes over time that affect both groups in a similar way. The interaction term *Russia* $\times$ *Post* indicates whether there is any post-invasion difference in the search interest between the groups. In some specifications, we include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

Dependent variables:					
Online Search Interest In:					
	USDT	USDT	USDT	USDT	USDT
Russia	0.003 (0.01)	0.003 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)
Post		0.21*** (0.01)	0.19*** (0.01)	0.19*** (0.01)	0.17*** (0.01)
Russia $\times$ Post			0.04*** (0.02)	0.04** (0.02)	0.04*** (0.02)
Constant	0.14*** (0.01)	0.09*** (0.005)	0.10*** (0.01)	0.10*** (0.01)	0.10*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.0002	0.53	0.54	0.54	0.29
Adjusted R ²	-0.001	0.53	0.54	0.54	0.28
	BTC	BTC	BTC	BTC	BTC
Russia	-0.23*** (0.01)	-0.23*** (0.01)	-0.24*** (0.01)	-0.24*** (0.01)	-0.24*** (0.01)
Post		-0.01 (0.01)	-0.04** (0.02)	-0.04** (0.02)	-0.04** (0.02)
Russia $\times$ Post			0.05** (0.02)	0.05** (0.02)	0.05** (0.02)
Constant	0.37*** (0.01)	0.38*** (0.01)	0.38*** (0.01)	0.38*** (0.01)	0.38*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.51	0.51	0.51	0.51	0.51
Adjusted R ²	0.51	0.51	0.51	0.51	0.51
	ETH	ETH	ETH	ETH	ETH
Russia	-0.17*** (0.01)	-0.17*** (0.01)	-0.17*** (0.01)	-0.17*** (0.01)	-0.17*** (0.01)
Post		0.06*** (0.01)	0.06*** (0.02)	0.06*** (0.02)	-0.06** (0.02)
Russia $\times$ Post			0.01 (0.03)	0.003 (0.03)	0.01 (0.03)
Constant	0.43*** (0.01)	0.41*** (0.01)	0.41*** (0.01)	0.41*** (0.01)	0.41*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.27	0.30	0.30	0.30	0.31
Adjusted R ²	0.27	0.30	0.30	0.29	0.30
	Binance	Binance	Binance	Binance	Binance
Russia	-0.01 (0.01)	-0.01 (0.01)	-0.001 (0.01)	-0.001 (0.01)	-0.002 (0.01)
Post		0.09*** (0.01)	0.11*** (0.01)	0.11*** (0.01)	0.08*** (0.02)
Russia $\times$ Post			-0.03 (0.02)	-0.02 (0.02)	-0.02 (0.02)
Constant	0.16*** (0.01)	0.14*** (0.01)	0.14*** (0.01)	0.14*** (0.01)	0.14*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.001	0.15	0.15	0.16	0.05
Adjusted R ²	-0.0003	0.15	0.15	0.15	0.03
	USD	USD	USD	USD	USD
Russia	-0.05*** (0.01)	-0.05*** (0.01)	-0.09*** (0.01)	-0.09*** (0.01)	-0.09*** (0.01)
Post		0.19*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.03* (0.02)
Russia $\times$ Post			0.14*** (0.02)	0.15*** (0.02)	0.15*** (0.02)
Constant	0.22*** (0.01)	0.17*** (0.01)	0.19*** (0.01)	0.19*** (0.01)	0.19*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	638	638	638	637	637
R ²	0.04	0.38	0.43	0.44	0.20
Adjusted R ²	0.03	0.38	0.43	0.43	0.19

Table A5: Long-Term DiD between Russia and the United States.

The table presents the results of the DiD regression analysis comparing the search interest in the selected keywords in Russia and the United States, before and after Russia's invasion of Ukraine on February 24, 2022. The dummy *Russia* is equal to one for observations related to Russian search interest; otherwise, it is zero. This variable captures systematic differences between Russia and the United States. The dummy variable *Post* is equal to one for all observations after February 24, 2022. The interaction term *Russia* $\times$ *Post* captures whether there is any post-invasion difference in search interest between the two groups. In some specifications, we include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

<i>Dependent variables:</i>					
Online Search Interest In:					
	USDT	USDT	USDT	USDT	USDT
Russia	0.08*** (0.004)	0.08*** (0.004)	0.002 (0.01)	0.002 (0.01)	0.002 (0.01)
Post		0.06*** (0.004)	-0.01 (0.01)	-0.01 (0.01)	-0.0004 (0.01)
Russia $\times$ Post			0.13*** (0.01)	0.13*** (0.01)	0.13*** (0.01)
Constant	0.09*** (0.003)	0.05*** (0.004)	0.09*** (0.004)	0.09*** (0.004)	0.09*** (0.004)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,268	1,267	1,267
R ²	0.22	0.33	0.47	0.47	0.43
Adjusted R ²	0.22	0.33	0.46	0.47	0.43
	BTC	BTC	BTC	BTC	BTC
Russia	-0.03*** (0.01)	-0.03*** (0.01)	-0.06*** (0.01)	-0.06*** (0.01)	-0.06*** (0.01)
Post		-0.08*** (0.01)	-0.10*** (0.01)	-0.10*** (0.01)	-0.07*** (0.01)
Russia $\times$ Post			0.04*** (0.01)	0.04*** (0.01)	0.04*** (0.01)
Constant	0.14*** (0.004)	0.18*** (0.005)	0.20*** (0.01)	0.20*** (0.01)	0.20*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,268	1,267	1,267
R ²	0.03	0.17	0.18	0.18	0.06
Adjusted R ²	0.02	0.17	0.18	0.18	0.05
	ETH	ETH	ETH	ETH	ETH
Russia	-0.07*** (0.01)	-0.07*** (0.01)	-0.14*** (0.01)	-0.14*** (0.01)	-0.14*** (0.01)
Post		-0.11*** (0.01)	-0.15*** (0.01)	-0.15*** (0.01)	-0.24*** (0.01)
Russia $\times$ Post			0.10*** (0.01)	0.10*** (0.01)	0.10*** (0.01)
Constant	0.18*** (0.005)	0.25*** (0.01)	0.28*** (0.01)	0.28*** (0.01)	0.28*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,268	1,267	1,267
R ²	0.01	0.20	0.24	0.25	0.23
Adjusted R ²	0.01	0.20	0.24	0.24	0.23
	Binance	Binance	Binance	Binance	Binance
Russia	0.04*** (0.005)	0.04*** (0.005)	0.01* (0.01)	0.01* (0.01)	0.01* (0.01)
Post		-0.06*** (0.005)	-0.08*** (0.01)	-0.08*** (0.01)	-0.06*** (0.01)
Russia $\times$ Post			0.05*** (0.01)	0.05*** (0.01)	0.05*** (0.01)
Constant	0.07*** (0.003)	0.11*** (0.004)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,268	1,267	1,267
R ²	0.06	0.17	0.19	0.19	0.11
Adjusted R ²	0.06	0.16	0.18	0.19	0.10
	USD	USD	USD	USD	USD
Russia	-0.24*** (0.01)	-0.24*** (0.01)	-0.12*** (0.01)	-0.12*** (0.01)	-0.12*** (0.01)
Post		0.20*** (0.01)	0.29*** (0.01)	0.29*** (0.01)	0.22*** (0.01)
Russia $\times$ Post			-0.19*** (0.01)	-0.19*** (0.01)	-0.19*** (0.01)
Constant	0.40*** (0.01)	0.28*** (0.01)	0.23*** (0.01)	0.23*** (0.01)	0.23*** (0.01)
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,268	1,267	1,267
R ²	0.41	0.67	0.73	0.73	0.66
Adjusted R ²	0.41	0.67	0.73	0.73	0.66

Table A6: Long-Term DiD between Russia and Ukraine.

The table presents the results of the DiD regression analysis comparing the search interest in the selected keywords in Russia and Ukraine, before and after Russia's invasion of Ukraine on February 24, 2022. The dummy *Russia* is equal to one for observations related to Russian search interest; otherwise, it is zero. This variable captures systematic differences between Russia and Ukraine. The dummy variable *Post* is equal to one for all observations after February 24, 2022. The interaction term *Russia* $\times$ *Post* captures whether there is any post-invasion difference in search interest between the two groups. In some specifications, we include controls for year- and month-fixed effects and for daily changes in RUB/USD exchange rate, S&P 500 index, VIX index, gold and oil prices. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses.

<i>Dependent variables:</i>					
Online Search Interest In:					
	USDT	USDT	USDT	USDT	USDT
Russia	0.004 (0.01)	0.004 (0.004)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)
Post		0.12*** (0.004)	0.11*** (0.01)	0.11*** (0.01)	0.11*** (0.01)
Russia $\times$ Post			0.02* (0.01)	0.02* (0.01)	0.02* (0.01)
Constant	0.17*** (0.004)	0.09*** (0.004)	0.10*** (0.005)	0.10*** (0.005)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,268	1,267	1,267
R ²	0.0004	0.36	0.36	0.36	0.11
Adjusted R ²	-0.0004	0.36	0.36	0.36	0.10
	BTC	BTC	BTC	BTC	BTC
Russia	-0.20*** (0.01)	-0.20*** (0.01)	-0.24*** (0.01)	-0.24*** (0.01)	-0.24*** (0.01)
Post		-0.09*** (0.01)	-0.12*** (0.01)	-0.12*** (0.01)	-0.10*** (0.01)
Russia $\times$ Post			0.07*** (0.01)	0.07*** (0.01)	0.07*** (0.01)
Constant	0.31*** (0.004)	0.36*** (0.01)	0.38*** (0.01)	0.38*** (0.01)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,268	1,267	1,267
R ²	0.48	0.57	0.58	0.58	0.56
Adjusted R ²	0.48	0.57	0.58	0.58	0.56
	ETH	ETH	ETH	ETH	ETH
Russia	-0.10*** (0.01)	-0.10*** (0.01)	-0.17*** (0.01)	-0.17*** (0.01)	-0.17*** (0.01)
Post		-0.12*** (0.01)	-0.17*** (0.01)	-0.17*** (0.01)	-0.24*** (0.02)
Russia $\times$ Post			0.12*** (0.02)	0.12*** (0.02)	0.12*** (0.01)
Constant	0.31*** (0.01)	0.38*** (0.01)	0.41*** (0.01)	0.41*** (0.01)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,268	1,267	1,267
R ²	0.11	0.25	0.28	0.28	0.25
Adjusted R ²	0.11	0.24	0.28	0.28	0.24
	Binance	Binance	Binance	Binance	Binance
Russia	-0.03*** (0.005)	-0.03*** (0.005)	-0.001 (0.01)	-0.001 (0.01)	-0.001 (0.01)
Post		-0.01*** (0.005)	0.01 (0.01)	0.01 (0.01)	0.02** (0.01)
Russia $\times$ Post			-0.04*** (0.01)	-0.04*** (0.01)	-0.04*** (0.01)
Constant	0.14*** (0.003)	0.15*** (0.004)	0.14*** (0.01)	0.14*** (0.01)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,268	1,267	1,267
R ²	0.02	0.03	0.04	0.05	0.05
Adjusted R ²	0.02	0.03	0.04	0.04	0.04
	USD	USD	USD	USD	USD
Russia	-0.03*** (0.01)	-0.03*** (0.01)	-0.09*** (0.01)	-0.09*** (0.01)	-0.09*** (0.01)
Post		0.06*** (0.01)	0.02** (0.01)	0.02** (0.01)	-0.04*** (0.01)
Russia $\times$ Post			0.08*** (0.01)	0.09*** (0.01)	0.09*** (0.01)
Constant	0.20*** (0.004)	0.16*** (0.01)	0.19*** (0.01)	0.19*** (0.01)	
Y/M Fixed Effects	No	No	No	No	Yes
Controls	No	No	No	Yes	Yes
Observations	1,268	1,268	1,268	1,267	1,267
R ²	0.03	0.11	0.14	0.15	0.08
Adjusted R ²	0.02	0.10	0.14	0.14	0.07